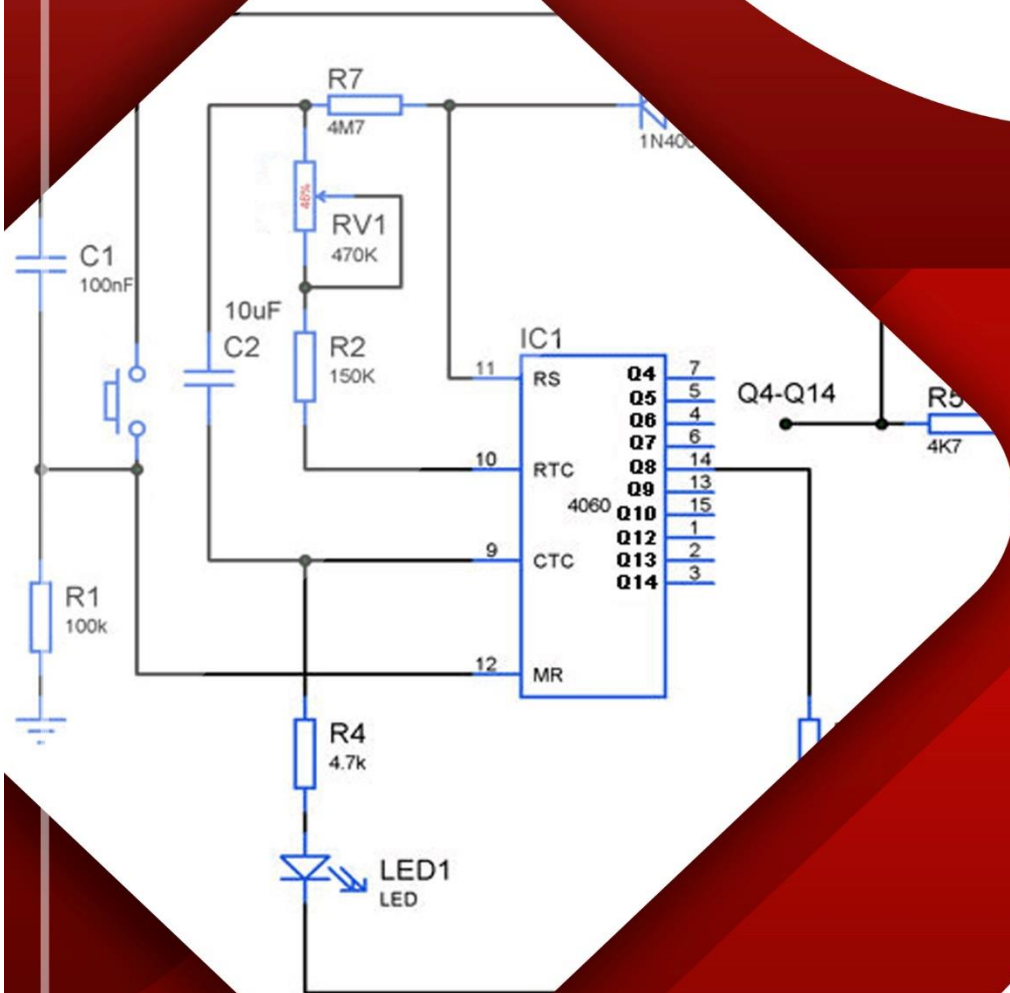


LABORATORY MANUAL IN **CIRCUIT 1**



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SAFETY

Safety in the electrical laboratory, as everywhere else, is a matter of the knowledge of potential hazards, following safety precautions, and common sense. Observing safety precautions is important due to pronounced hazards in any electrical/computer engineering laboratory. Death is usually certain when 0.1 ampere or more flows through the head or upper thorax and have been fatal to persons with coronary conditions. The current depends on body resistance, the resistance between body and ground, and the voltage source. If the skin is wet, the heart is weak, the body contact with ground is large and direct, then 40 volts could be fatal. Therefore, never take a chance on "low" voltage. When working in a laboratory, injuries such as burns, broken bones, sprains, or damage to eyes are possible and precautions must be taken to avoid these as well as the much less common fatal electrical shock. Make sure that you have handy emergency phone numbers to call for assistance if necessary. If any safety questions arise, consult the lab demonstrator or technical assistant/technician for guidance and instructions. Observing proper safety precautions is important when working in the laboratory to prevent harm to yourself or others. The most common hazard is the electric shock which can be fatal if one is not careful.

Acquaint yourself with the location of the following safety items within the lab.

- a. fire extinguisher
- b. first aid kit
- c. telephone and emergency numbers

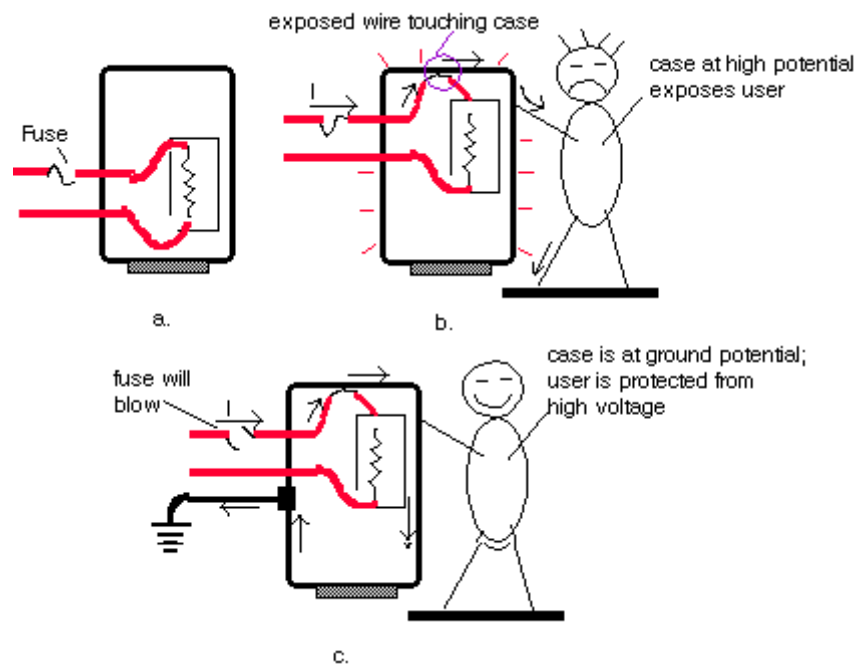
Electric shock

Shock is caused by passing an electric current through the human body. The severity depends mainly on the amount of current and is less function of the applied voltage. The threshold of electric shock is about 1 mA which usually gives an unpleasant tingling. For currents above 10 mA, severe muscle pain occurs and the victim can't let go of the conductor due to muscle spasm. Current between 100 mA and 200 mA (50 Hz AC) causes ventricular fibrillation of the heart and is most likely to be lethal.

What is the voltage required for a fatal current to flow? This depends on the skin resistance. Wet skin can have a resistance as low as 150 Ohm and dry skin may have a resistance of 15 kOhm. Arms and legs have a resistance of about 100 Ohm and the trunk 200 Ohm. This implies that 240 V can cause about 500 mA to flow in the body if the skin is wet and thus be fatal. In addition skin resistance falls quickly at the point of contact, so it is important to break the contact as quickly as possible to prevent the current from rising to lethal levels.

Equipment grounding

Grounding is very important. Improper grounding can be the source of errors, noise and a lot of trouble. Here we will focus on equipment grounding as a protection against electrical shocks. Electric instruments and appliances have equipment casings that are electrically insulated from the wires that carry the power. The isolation is provided by the insulation of the wires as shown in the figure a below. However, if the wire insulation gets damaged and makes contact to the casing, the casing will be at the high voltage supplied by the wires. If the user touches the instrument he or she will feel the high voltage. If, while standing on a wet floor, a user simultaneously comes in contact with the instrument case and a pipe or faucet connected to ground, a sizable current can flow through him or her, as shown in Figure b. However, if the case is connected to the ground by use of a third (ground) wire, the current will flow from the hot wire directly to the ground and bypass the user as illustrated in figure c.



Equipments with a three wire cord is thus much safer to use. The ground wire (3rd wire) which is connected to metal case is also connected to the earth ground (usually a pipe or bar in the ground) through the wall plug outlet.

Always observe the following safety precautions when working in the laboratory:

1. Do not work alone while working with high voltages or on energized electrical equipment or electrically operated machinery like a drill.

2. Power must be switched off whenever an experiment or project is being assembled, disassembled, or modified. Discharge any high voltage points to grounds with a well insulated jumper. Remember that capacitors can store dangerous quantities of energy.
3. Make measurements on live circuits or discharge capacitors with well insulated probes keeping one hand behind your back or in your pocket. Do not allow any part of your body to contact any part of the circuit or equipment connected to the circuit.
4. After switching power off, discharge any capacitors that were in the circuit. Do not trust supposedly discharged capacitors. Certain types of capacitors can build up a residual charge after being discharged. Use a shorting bar across the capacitor, and keep it connected until ready for use. If you use electrolytic capacitors, do not :
 - put excessive voltage across them
 - put ac across them
 - connect them in reverse polarity
5. Take extreme care when using tools that can cause short circuits if accidental contact is made to other circuit elements. Only tools with insulated handles should be used.
6. If a person comes in contact with a high voltage, immediately shut off power. Do not attempt to remove a person in contact with a high voltage unless you are insulated from them. If the victim is not breathing, apply CPR immediately continuing until he/she is revived, and have someone dial emergency numbers for assistance.
7. Check wire current carrying capacity if you will be using high currents. Also make sure your leads are rated to withstand the voltages you are using. This includes instrument leads.
8. Avoid simultaneous touching of any metal chassis used as an enclosure for your circuits and any pipes in the laboratory that may make contact with the earth, such as a water pipe. Use a floating voltmeter to measure the voltage from ground to the chassis to see if a hazardous potential difference exists.
9. Make sure that the lab instruments are at ground potential by using the ground terminal supplied on the instrument. Never handle wet, damp, or ungrounded electrical equipment.
10. Never touch electrical equipment while standing on a damp or metal floor.
11. Wearing a ring or watch can be hazardous in an electrical lab since such items make good electrodes for the human body.
12. When using rotating machinery, place neckties or necklaces inside your shirt or, better yet, remove them.
13. Never open field circuits of D-C motors because the resulting dangerously high speeds may cause a "mechanical explosion".

14. Keep your eyes away from arcing points. High intensity arcs may seriously impair your vision or a shower of molten copper may cause permanent eye injury.
15. Never operate the black circuit breakers on the main and branch circuit panels.
16. In an emergency all power in the laboratory can be switched off by depressing the large red button on the main breaker panel. Locate it. It is to be used for emergencies only.
17. Chairs and stools should be kept under benches when not in use. Sit upright on chairs or stools keeping the feet on the floor. Be alert for wet floors near the stools.
18. Horseplay, running, or practical jokes must not occur in the laboratory.
19. Never use water on an electrical fire. If possible switch power off, then use CO₂ or a dry type fire extinguisher. Locate extinguishers and read operating instructions before an emergency occurs.
20. Never plunge for a falling part of a live circuit such as leads or measuring equipment.
21. Never touch even one wire of a circuit; it may be hot.
22. Avoid heat dissipating surfaces of high wattage resistors and loads because they can cause severe burns.
23. Keep clear of rotating machinery.

Precautionary Steps Before Starting an Experiment so as Not to Waste Time Allocated

- a) Read materials related to experiment before hand as preparation for pre-lab quiz and experimental calculation.
- b) Make sure that apparatus to be used are in good condition. Seek help from technicians or the lab demonstrator in charge should any problem arises.
 - Power supply is working properly i.e. $I_{\max}$ (maximum current) LED indicator is disable. Maximum current will retard the dial movement and eventually damage the equipment. Two factors that will light up the LED indicator are short circuit and insufficient supply of current by the equipment itself. To monitor and maintain a constant power supply, the equipment must be connected to circuit during voltage measurement. DMM are not to be used simultaneously with oscilloscope to avert wrong results.
 - Digital multimeter (DMM) with low battery indicated is not to be used. By proper connection, check fuses functionality (especially important for current measurement). Comprehend the use of DMM for various functions. Verify

measurements obtained with theoretical values calculated as it is quite often where 2 decimal point reading and 3 decimal point reading are very much deviated.

- The functionality of voltage waveform generators are to be understood. Make sure that frequency desired is displayed by selecting appropriate multiplier knob. Improper settings (ie selected knob is not set at minimum (in direction of CAL – calibrate) at the bottom of knob) might result in misleading values and hence incorrect results. Avoid connecting oscilloscope together with DMM as this will lead to erroneous result.
 - Make sure both analog and digital oscilloscopes are properly calibrated by positioning sweep variables for VOLT / DIV in direction of CAL. Calibration can also be achieved by stand alone operation where coaxial cable connects CH1 to bottom left hand terminal of oscilloscope. This procedure also verifies coaxial cable continuity.
- c) Internal circuitry configuration of breadboard or Vero board should be at students' fingertips (i.e. holes are connected horizontally not vertically for the main part with engravings disconnecting in-line holes).
- d) Students should be rest assured that measured values (theoretical values) of discrete components retrieved i.e. resistor, capacitor and inductor are in accordance the required ones.
- e) Continuity check of connecter or wire using DMM should be performed prior to proceeding an experiment. Minimize wires usage to avert mistakes.
- f) It is unethical and wrong for students to falsify results as to make them appear exactly consistent with theoretical calculations.

Table of Contents

Experiment 1.....	Introduction to experiments
Experiment 2.....	Resistor color code and measurement of resistance
Experiment 3.....	Voltage and current measurements in an electric circuit
Experiment 4.....	.ohm's law
Experiment 5.....	Power in dc circuits
<i>Experiment 6.....</i>	<i>Series resistive circuits</i>
Experiment 7.....	Parallel resistive circuits
Experiment 8.....	Serries parallel circuit
Experiment 9.....	Kirchhoff's voltage law
Experiment 10.....	Voltage divider
Experiment 11.....	Kirchhoff's Current law
Experiment 13.....	current divider
Experiment 14.....	Electronics workbench
Experiment 15.....	Superposition
Experiment 16.....	Thevenin theorem
Experiment 17.....	Maximum power transfer
Experiment 18.....	Capacitor charging and discharging
Experiment 19.....	Rate of Change of Inductor

INTRODUCTION TO EXPERIMENTS

Experiment # 1

OBJECTIVES

1. To become familiar with basic laboratory equipment and components
2. To read and interpret electrical and electronic drawings
3. To gather and tabulate data in the laboratory

BASIC INFORMATION

The electricity/electronics laboratory is a place of learning, whether it be an industrial, commercial, or school laboratory. The experiments support the overall learning process by helping you do the following:

- A. Develop practical skills in the use of power supplies, instruments, and components in a hands-on test environment
- B. Develop the techniques of gathering accurate experimental information and communicating that information to others through written reports and tabulated data

Although specific objectives will be described in the course of the experiments that follow, the preceding two items are the keys to all of them. To build the circuits, perform the experiments, gather and interpret data, and report your conclusions you will need to be familiar with the following basic equipment, tools, and techniques of the electricity/electronics laboratory:

1. Schematic diagrams and symbols
2. Power sources and supplies
3. Instruments
4. Tables and reports

Schematic Diagrams and Symbols

Schematic diagrams are a form of language that helps us communicate information about the electrical relationships in a circuit. Schematic diagrams do not indicate physical positions or sizes of components, nor do they show actual points of interconnection (though certain drawing conventions do try to indicate all these in somewhat simplified form). The symbols used in schematic diagrams represent the components and conductors of the circuit,

but again, the symbols do not try to show the physical shape or size of the actual component. They do in some respects convey the idea of a characteristic feature of the component. Figure 1-1 shows some of the graphic symbols used in this book. When specialized or unique symbols are used in schematic diagrams, they are labeled and identified.

Power Sources and Supplies

Direct Current. The two basic power sources used in the experiments in this book are direct current and alternating current. Direct current can be supplied by electric cells such as the familiar D, C, and AA cells used in many portable electronic devices. Batteries consist of combinations of cells designed to produce voltages that cannot be supplied by a single cell. Automobile batteries and the small 9-V batteries used in portable radios and wireless alarm systems actually contain combinations of cells. Although cells and batteries are convenient and widely used in the laboratory, especially to power portable instruments, they do not produce a constant voltage and they are not easy to use over a wide range of voltages. For such purposes a variable voltage-regulated supply is most often used.

A variable voltage-regulated power supply is one that can be manually adjusted to deliver any required voltage within its range of operation. The term *voltage-regulated* means that the voltage delivered by the supply remains constant despite changes in load current, within specified limits. Some power supplies are designed to provide two or more independent dc voltages, in which case the instrument has separate controls and separate output terminals.

The polarity of the dc terminals on the supply is usually marked either -, +, or GND and V+. By convention a red jack is used for the positive terminal, and a black jack is used for the negative terminal of the supply.

Alternating Current. Alternating current sources include the wall outlet supplying a nominal 120 V as well as variable voltage-regulated supplies similar to those for dc. Single-voltage ac sources are also common for particular power requirements, such as telephone answering machines and video games. Variable-voltage autotransformers are also widely used. Although the output of these supplies is usually unregulated, it is constant enough for most laboratory applications.

Using a Power Supply. The following general precautions should be observed in using a power supply:

1. Read the operating manual carefully and be certain that you understand the instructions before turning the power supply on for the first time.
2. Never create a short circuit across the output terminals, or you may damage the supply. Keep the leads connected to the output terminals from making contact with each other.
3. If any component on the experimental circuit appears to be overheating after power is applied, turn the supply off and determine the cause before you proceed with the experiment.
4. To prevent damage to the power supply, do not switch the supply on and off excessively. If an experiment requires power to be interrupted frequently, use an external switch on the breadboard to apply and remove power from the circuit.
5. Do not operate the supply beyond its *rated* current capacity. If an ammeter indicates that you are exceeding the current capability of the power supply, turn it off and check the experimental circuit to determine why it is drawing excessive current.
6. Always be alert to shock hazards. Even equipment that has what may appear to be relatively safe voltages can produce dangerous if not lethal currents under certain conditions.

Instruments

Electrical and electronic instruments are used in the laboratory to measure quantities, to provide external signal stimuli to circuits, and to view graphically the behavior of circuits.

Measuring Instruments. The two types of measuring instruments in common use are analog meters and digital meters. Analog meters can be further classified as electronic meters or electromechanical meters. In the analog meter (both electronic and electromechanical) a pointer indicates the measured value on a calibrated scale. This usually requires making an accurate estimate of the exact position of the pointer on the scale. The value measured by a digital meter appears as a number consisting of three or four digits on a lighted display.

Anyone who can read numbers can therefore read the quantity being measured by a digital meter. Voltage measurements are made primarily with an instrument called a *voltmeter*. Other instruments may also be used to measure voltage— for example, the oscilloscope—but voltage measurement is not the

Current is measured with an *ammeter*. If the ammeter is designed to measure very small quantities, it is known specifically as a *milliammeter* or *microammeter*.

Resistance is measured with an *ohmmeter*. Whereas the voltmeter and ammeter are always used in circuits to which power has been applied, ohmmeters are always used in unpowered circuits. The ohmmeters carry their own power supply, usually in the form of one or two 1.5-V cells. Another common laboratory meter, the *wattmeter*, measures power.

Meters may be single-function devices, such as the voltmeter, which can measure only voltage. Or they may be multipurpose devices, such as the multimeter, which is used to measure a variety of electrical quantities, such as voltage, current, and resistance.

Signal Generators. Many test procedures and experiments require application of voltages with frequencies other than the 60 Hz supplied by the power companies. In addition, certain applications call for alternating current with waveforms other than a sine wave. Signal generators and function generators are designed to produce variable-frequency outputs as well as sine, sawtooth, and square waves. The voltage output of these generators is not intended to be a continuous power source. Rather, the outputs provide signal stimuli for circuits so that the behavior of the circuit may be observed under these signal conditions. For example, a signal generator may simulate the modulated radio wave that enters the antenna of a radio receiver in order to troubleshoot various circuits of the radio. The signal generator can produce precise, stable signals, but it does not measure frequencies or voltages.

Oscilloscope. Modern oscilloscopes are used to observe circuit behavior graphically by displaying voltage waveforms on the screen of a cathode-ray tube. In addition to making it possible to view the shape of the wave, scopes can also be used to measure the voltage and frequency of the wave. If more than one waveform is present, the oscilloscope also permits determining the phase displacement between the waves. Dual-trace oscilloscopes make it possible to view two waveforms simultaneously on the same screen.

The Internet

An electronics technician requires many sources of information, along with technical and communications skills, to properly perform on the job. Information covering new product information, service bulletins, component specifications, training and workshops, and professional organizations is vital to keep up with current technology. The Internet, with its massive amounts of online information, can be used to provide the necessary technical resources.

Information dealing with professional organizations and electronics careers can be informative and helpful as you begin your study of electronics.

Tables and Reports

The purpose of most experiments is to observe reactions, measure quantities, record data, analyze the data, and, if possible, draw some conclusions. Attention to the following suggestions will result in more effective laboratory work:

1. Because gathering data is an important element of the experimental process, it must be done carefully and systematically.
2. The data should be tabulated in a logical fashion. The tables of data should be legible, the units of each quantity identified, and the level of accuracy consistent for like quantities.
3. Accuracy in reading meters and adjusting voltages of power supplies is essential if the data are to have any credibility. If special conditions arise or particular results are distorted, the facts and circumstances should be noted along with the tabulated data.
4. Many tables require calculations based on circuit analysis and measured data. In tabulating calculated values it is important to be consistent with the number of significant figures in your answers.
5. Some circuits should be analyzed and their performance calculated before performing an experiment. The measured quantities can then be compared with the calculated values. Discrepancies should be investigated and discussed.
6. If graphs are required, appropriate scales and type of graph paper should be used so that the plotted graph clearly shows trends and the particular relationships being demonstrated in the experiment.

7. The tabulated data, graphs, calculations, and the like should lead to some conclusions about the principles or concepts that are central to the experiment.
8. The conclusion should be written neatly, stated clearly, and referred to the supporting experimental data and observations.
9. The technical report should be free of spelling and grammatical errors and use the technical terminology appropriate to the subject of the experiment.

SUMMARY

1. The basic equipment, tools, and techniques of the electricity/electronics laboratory are schematic diagrams and symbols, power sources and supplies, instruments, and tables and reports.
2. Schematic diagrams and symbols are the shorthand language of electrical and electronic circuits.
3. Schematic diagrams show the electrical relationships of components and conductors in a circuit.
4. Schematic diagrams do not show the actual physical locations or connections of a circuit. Symbols do not represent the actual physical shape or size of the components.
5. The two main power systems are direct current and alternating current.
6. Electric cells such as the familiar D, C, and AA cells provide a source of direct current for portable devices.
7. Most laboratory dc requirements are provided by variable voltage-regulated power supplies.
8. The most common ac source is the wall outlet that provides a nominal 230 V.
9. Alternating current sources include variable voltage-regulated power supplies and single-voltage supplies.
10. A popular ac source in the laboratory is the variable-voltage autotransformer. This type of power source is usually unregulated.
11. The term voltage-regulated means that the voltage will remain constant at a particular setting despite changes in load current, within specified limits.

12. Instruments in the electricity/electronics laboratory are used to measure quantities, provide external signal sources to circuits, and to view the behavior of circuits graphically.
13. Voltmeters, ammeters, and ohmmeters can be either analog or digital meters.
14. Analog meters can be either electronic or electromechanical.
15. Meters can be single-function instruments, such as voltmeters or ammeters, or they can be multipurpose devices in which the voltmeter, ammeter, and ohmmeter functions are in the same housing and use the same display.
16. Signal generators produce variable frequencies from 1 Hz to as many as hundreds of megahertz, although no single signal generator produces all these frequencies.
17. A signal generator generally produces a sine wave, but some manufacturers produce models that also provide a square-wave output.
18. Function generators are capable of providing variable frequency and amplitude sine wave, sawtooth, and square waveforms.
19. The oscilloscope is used to observe the shape, frequency, and amplitude of waveforms in a circuit. Dual-trace oscilloscopes are capable of displaying two sets of waves on the same screen simultaneously. Oscilloscopes are used to measure voltage, frequency, and phase displacement of ac waveforms.
20. An essential part of experimental work in the laboratory is the gathering and tabulating of data.
21. The tabulated data, graphs, calculations, and observations made during an experiment should lead to some conclusion about the principles and concepts involved in the experiment.
22. Written reports are used to communicate the data, observations, and conclusions of the experimenter to others.
23. The Internet is a source of abundant information useful to the electronics technician

MATERIALS REQUIRED

Power Supplies:

- Variable ac • Variable 0 -15 V dc, regulated
- Variable-voltage autotransformer (Variac or equivalent)

Instruments:

A varied selection of the following should be available:

DMM • Analog voltmeter (range not critical)

VOM • Analog ammeter (range not critical)

Miscellaneous:

- A group of components provided by the instructor (components should be mounted and labeled)
- Operating manuals for all instruments provided

A. Schematic Diagrams and Symbols

- A1. Your instructor will provide you with a group of components, only a few of which are pictured in the schematic diagram of Figure 1-2. In Table 1-1 identify those of your components that are represented in the figure. Next to the symbol number in the table, record the name of the component and sketch what the component actually looks like.

B. Power Sources and Supplies

- B1. Your instructor will give you a power supply of the type used in the laboratory. Obtain a copy of the operating manual for the supply.
- B2. Read the manual to become thoroughly familiar with the controls and terminals on the front panel of the power supply. At this point it is not necessary to become familiar with the actual operation of the supply.
- B3. Complete Table 1-2 Power Supply Controls and Features.

C. Instruments

- C1. Your instructor will give you two instruments with which you are to become familiar. Obtain a copy of the operating manuals for these two instruments.
- C2. Read the manuals carefully to become familiar with the various switches

and jacks on the front panel.

- C3. Complete Table 1-3(a) Meter 1 Functions and Features, and Table 1-3(b) Meter 2 Functions and Features.

Table 1-1 Schematic Diagram Symbols

#	Component Name	Sketch of Component		#	Component Name	Sketch of Component
1				4		
2				5		
3				6		

Table 1-2 Power Supply Controls and Features

Manufacturer_____	
Model no. _____	
Type of Power supply	
_____ DC	_____ Regulated
_____ AC	_____ Unregulated
_____ Single voltage(_____ V)	
_____ Multiple voltage (voltages: _____)	
_____ No. of output terminals _____	
Controls and Switches	Function

Table 1-3(a) Meter 1 Functions and Features

Manufacturer _____		
Model no. _____		
Type of meter _____		
Type of Power supply		
_____ DC	_____ Electronic	_____ Battery operated
_____ AC	_____ Analog	_____ Line Operated
_____ DC/AC	_____ Digital	_____ Battery/Line operated
_____ Single meter	_____ Multimeter	
Switches, Ranges, Terminals, Special Features	Function	

Table 1-3(b) Meter 2 Functions and Features

Manufacturer _____		
Model no. _____		
Type of meter _____		
Type of Power supply		
_____ DC	_____ Electronic	_____ Battery operated
_____ AC	_____ Analog	_____ Line Operated
_____ DC/AC	_____ Digital	_____ Battery/Line operated
_____ Single meter	_____ Multimeter	

Switches, Ranges, Terminals, Special Features	Function

Questions

1. Is it possible to use the same calibrated scale on the 3 – V as on the 300 – V range of a voltmeter? Explain.

[illegible]

2. Is the scale on your analog ohmmeter linear or nonlinear? Give examples to support your answer.

3. An unknown voltage (known only to be less than 600 V dc) is to be measured using a dc voltmeter having the following ranges: 600; 300; 15; 3. Explain how you would measure this voltage with your voltmeter.

4. What is meant by zeroing an ohmmeter? How is it done?

5. What is the difference between the readout of an analog electronic voltmeter and a digital voltmeter?

This image shows a single page of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page, leaving small margins at the top and bottom. There are no vertical margin lines, text, or other markings on the page.

6. List two advantages of a variable regulated dc power over cells and batteries.

[illegible]

7. Select two features of the power supply investigated in Table 1-2 and describe why they are particularly useful for laboratory experiments.

[illegible]

8. Select two features of the meters investigated in Tables 1-3(a) and (b) and describe why these features are particularly useful for laboratory experiments.

[illegible]

Conclusions:

RESISTOR COLOR CODE AND MEASUREMENT OF RESISTANCE
Experiment # 2

Introduction

The experiment on "Resistor Color Code and Measurement of Resistance" is a fundamental exercise in the field of electronics and electrical engineering. This experiment introduces the basics of identifying resistor values through their color bands, a skill essential for any work involving electronic circuits. Resistors are ubiquitous components in electronic devices, functioning to limit current flow, adjust signal levels, divide voltages, and much more.

The color-coding system on resistors consists of several bands that represent numbers and multipliers to indicate the resistor's value and tolerance. Learning to interpret these color codes is a fundamental skill for anyone involved in circuit design or assembly. Additionally, the experiment involves using tools like multimeters to measure the resistance of these components, allowing for a practical understanding of how theoretical values translate into real-world measurements.

Learning Objectives

1. **Understand Resistor Color Codes:** Learn to identify resistor values accurately by interpreting the color bands on resistors. This includes understanding the significance of each band and the color code chart.
2. **Develop Precision in Reading Color Codes:** Enhance skills in accurately reading color codes, given that some colors can be difficult to distinguish and that errors in reading these codes can lead to circuit errors.
3. **Measurement of Resistance Using Tools:** Gain proficiency in using tools such as multimeters to measure the resistance of resistors. This includes understanding how to properly connect the multimeter to the resistor and

interpret the readings.

4. Compare Theoretical and Practical Values: Learn to compare the color-coded theoretical value of a resistor with its practical measured value, understanding the reasons for any discrepancies, such as manufacturing tolerances.
5. Understanding Tolerance and its Importance: Grasp the concept of tolerance as indicated by the color bands and why it is crucial in selecting resistors for specific circuit applications.
6. Circuit Application Knowledge: Develop an understanding of how resistors function in circuits and the importance of selecting the correct resistance values for specific applications.
7. Safety Practices in Handling Electronic Components: Learn safe handling practices for electronic components and the use of measurement tools to prevent damage to components and equipment.
8. Interpretation of Results and Error Analysis: Enhance skills in interpreting results, conducting error analysis, and understanding the practical limitations and variations in electronic components.

By mastering these objectives, students will not only learn the basics of resistor identification and measurement but also lay a strong foundation for more advanced studies and applications in electronics and electrical engineering.

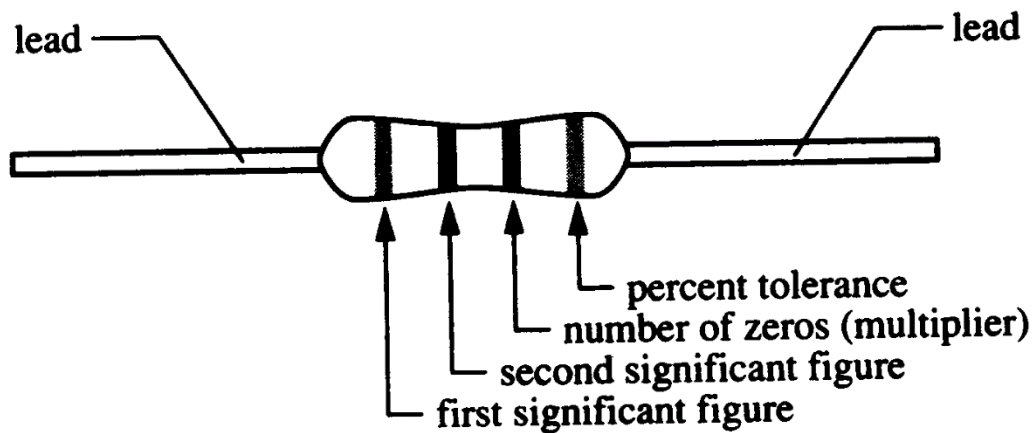
The *ohm* is the unit of resistance, and it is represented by the symbol Ω (Greek letter omega). Resistance values are indicated by a standard color code that manufacturers have adopted. This code uses color bands on the body of the resistor. The colors and their numerical values are given in the resistor color chart, Table 2-1. This code is used for 1/8 W, 1/4 W, 1/2 W, 1-W, and 2-W resistors.

The basic resistor is shown in Figure 2-1. The standard color-code marking consists of four bands around the body of the resistor. The color of the first band indicates the first significant figure of the resistance value. The second band indicates the second significant figure. The color of the third band indicates the number of zeros that follow the first two significant figures. If the third band is gold or silver, the resistance value is less than 10 ohms. For resistors less than 10 ohms, the third band indicates a fractional value of the first two significant figures. Table 2-1 shows the resistor color codes.

Table 2-1 Resistor Color Codes

Color Band	Significant Digit (First and Second Band)	Multiplier (Number of Zeroes)	Tolerance	Failure Rate (Fifth Band)
Black	0	1	+/- 20%	-
Brown	1	10	+/- 1%	1.0
Red	2	100	+/- 2%	0.1
Orange	3	1000	+/- 3%	0.01
Yellow	4	10000	+/- 4%	0.001
Green	5	100000	Not Used	Not Used
Blue	6	1000000		
Violet	7	10000000		
Gray	8	Not Used		
White	9			
Gold	Not Used		+/- 5%	
Silver			+/- 10%	
No color band		+/- 20%		

Figure 2.1 Resistor Color Code



The fourth band indicates the *percent tolerance* of the resistance. Percent tolerance is the amount the resistance may vary from the value indicated by the color code. Because resistors are mass produced, variations in materials will affect their actual resistance. Many circuits can still operate as designed even if the resistors in the circuit do not have the precise value specified. Tolerances are usually given as plus or minus the nominal, or color-code, value.

High-precision resistors have five bands. The first three bands indicate the first three significant figures of the resistance; the fourth band indicates the number of zeros; fifth band is the percent tolerance.

Resistors manufactured to military specification also contain a fifth band. The fifth band in this case is used to evaluate the reliability. The figure given is the percentage of defective parts per 1000 hours of operation.

Wirewound, high-wattage resistors usually are not color coded but have the resistance value and wattage printed on the body of the resistor.

To avoid having to write all the zeros for high value resistors the metric abbreviations of k (for 1000) and M for (1000) are used. For example 33,000 Ω can be written as 33 k Ω

(pronounced 33 kay, or 33 kilohms). 1,200,000 Ω can be written as 1.2 M Ω (pronounced 1.2 meg, or 1.2 megohms).

Materials Required

<i>Instruments:</i>	Analog multimeter
<i>Resistors:</i>	Assorted resistance values and tolerances; ½ W, color coded 10 k Ω potentiometer
<i>Miscellaneous:</i>	hookup wire about 12 in. long Wire cutters, crayons

Procedure

1. Your instructor will give you 10 resistors of various values and tolerances. Examine each one and determine its resistance and tolerance according to its color code. In Table 2-3, using crayons, color the cell with the corresponding shade of the four color bands of each resistor. On the same table, record the coded resistance value, and tolerance.
2. Refer to experiment 1 and the operator's manuals on the use of an analog multimeter to measure resistance. Zero the ohmmeter. Using the coded resistance value as a guide, select an appropriate meter range and measure the resistance of the 10 resistors. Record your readings under "Measured Value" in Table 2-3.
3. For each of the measured resistance values in step 2, calculate the percent error using the equation:

$$\% \text{Accuracy} = \frac{\text{Coded Value} - \text{Measured Value}}{\text{Coded Value}} \times 100$$

Record this % error in Table 2-3

- 4.a. Measure and record the resistance of a short length of hookup wire. R = _____ Ω
- 4.b. Select one of the resistance in step 1 and connect the wire in step 4a across it as shown in the figure. By connecting the wires across the leads of the resistor, the resistor has been short circuited. Measure the resistance across the resistor-hookup wire combination. R = _____ Ω

5. Connect the hookup wire across the leads of your ohmmeter. Note the reading of the meter (you need not record this reading). Using your wire cutter, cut the wire cutter in half; the result is an open circuit. Note the reaction of the meter when the wire is cut.
- 6.a. Examine the potentiometer given to you. Place it on the lab table with the shaft pointing up. The terminals of your potentiometer will be A,B, and C. Rotate the shaft completely counter clockwise (CCW). Connect the ohmmeter leads to terminals A and B and measure the resistance between terminals A and B (R_{AB}) Record your reading in Table 2-4.
- 6.b. Next, measure the resistance between terminals A and C (R_{AC}). Record your reading in Table 2-4.
- 6.c. Finally measure the resistance between terminals B and C (R_{BC}) and record the reading in Table 2-4.
- 7.a. With the ohmmeter connected to terminals B and C, turn the shaft completely clockwise (CW). Note the reaction of the ohmmeter as the shaft is turned. Measure the resistance between terminals B and C and record it in Table 2-4,
- 7.b. Connect the ohmmeter across terminals A and C. Measure the resistance between terminals A and C record it in Table 2-4.
- 7.c. Measure the resistance between terminals A and B and record it in Table 2-4.
8. Calculate the value of $R_{AC} + R_{BC}$ for steps 6 and 7 and record in Table 2-4
9. Connect a length of hookup wire from terminal B to terminal C. Measure the resistance between terminals A and C. $R_{AC} = ______ \Omega$. Disconnect the hookup wire.
10. Connect the hookup wire from terminal A and C. Measure the resistance between terminals B and C. $R_{BC} = ______ \Omega$. Disconnect the hookup wire.

Table 2-3. Resistor Measured Resistance versus Color-Coded Vaues

	Resitors									
	1	2	3	4	5	6	7	8	9	10
First Color band										
Second Color band										
Third Color band										

Fourth Color band										
Coded Value										
Tolerance, %										
Measured Value										
error (%)										

Table 2-4. Potentiometer Measurements

Step	Setting of Potentiometer Control	R_{AB}, Ω	R_{AC}, Ω	R_{BC}, Ω	$R_{AC} + R_{BC}$ Calculated Value
6	CCW				
7	CW				
9	$R_{AC} =$ _____				
10	$R_{BC} =$ _____				

Questions

1. When using an analog multimeter, at which end of the ohms scale are the resistance measurements more reliable, the zero end or the infinity end? Explain

2. Complete the table below by supplying color code for each of the following coded value of the carbon resistors?

Coded value	1 st band	2 nd band	3 rd band	4 th band
0.27 Ω , 1/2 W, 5%				
2.2 Ω , 1/4 W, 10%				
39 Ω , 1/8 W, 10%				
560 Ω , 1/2 W, 5%				
33 k Ω , 1 W, 5%				

3. Can inserting a resistor in a circuit produce an effect similar to a short circuit?
Explain.

4. Can inserting a resistor in a circuit produce an effect similar to a open circuit?
Explain.

5. Explain the significance of the reaction noted in step 4b of the procedure.

6. An ohmmeter has the following ranges: R x 1, R x 10, R x 100, R x 1k. at the R x 10 range, the pointer indicates 1500Ω . If the range were changed to R x 1, would the pointer move to zero end or toward the infinity end? Would the new measurement be more or less accurate than at the R x 10 range? Explain

CONCLUSION:

VOLTAGE AND CURRENT MEASUREMENTS IN AN ELECTRIC CIRCUIT

Experiment # 3

Introduction

The experiment on "Voltage and Current Measurements in an Electric Circuit" is a crucial part of learning in the field of electrical engineering and electronics. This experiment is designed to provide a hands-on understanding of how to accurately measure voltage and current, two fundamental parameters in any electrical circuit. Voltage, often described as the electric potential difference, is the force that drives electric current through a circuit. Current, on the other hand, is the flow of electric charge through a conductor.

Understanding how to measure these two parameters accurately is essential for designing, testing, and troubleshooting electrical circuits. The experiment typically involves setting up simple circuits and using measurement tools like voltmeters and ammeters. These instruments are used to measure voltage across components and current through components, respectively. The experiment provides practical experience that complements theoretical knowledge, enabling a deeper understanding of how circuits operate in real-world scenarios.

Learning Objectives

1. Fundamental Understanding of Voltage and Current: Develop a solid understanding of what voltage and current are, including their units of measurement (volts for voltage and amperes for current).
2. Proficiency in Using Measurement Tools: Gain hands-on experience in using voltmeters and ammeters, understanding how to connect them correctly in a circuit (voltmeters in parallel and ammeters in series).

3. Circuit Assembly and Analysis: Learn how to assemble basic circuits and perform measurements, reinforcing the understanding of circuit principles like Ohm's Law and Kirchhoff's Circuit Laws.
4. Accurate Measurement Techniques: Develop the skill to take accurate measurements, understanding the importance of precision in voltage and current measurements for circuit analysis.
5. Interpreting and Recording Data: Enhance the ability to interpret the data obtained from measurements, record it accurately, and use it to analyze the circuit's behavior.
6. Understanding the Impact of Measurement on Circuits: Learn about the impact that measurement tools can have on a circuit (like the loading effect) and how to minimize it.
7. Safety Practices in Electrical Measurements: Understand and apply safety practices while handling electrical circuits and measurement instruments to prevent accidents and damage to equipment.
8. Troubleshooting Skills: Develop basic troubleshooting skills by identifying and rectifying common issues in circuit setups that can lead to inaccurate measurements.
9. Real-World Application Awareness: Recognize the real-world applications of these measurement techniques in various fields of electronics and electrical engineering.

By achieving these objectives, students will be equipped with the essential skills and knowledge required to measure and analyze electrical circuits effectively, a critical competency in the field of electronics and electrical engineering.

MATERIALS REQUIRED:

Power supply: dc power supply 0-10 V

Instruments: 2 analog or digital multimeter
Resistor: 7.5 k Ω $\pm$ 5 percent

breadboard
connecting wires

Information about voltage and current values in an electronic circuit is crucially important in the determination of faulty components. Measurement of voltage, in particular, is the most common activity in a technician's daily routine. Several different kinds of voltage- and/or current-measuring instruments are available. The Laboratory has both analog and digital multimeters for your use.

This experiment will familiarize you with basic voltage and current measurements across and through a resistor that is connected to a dc power supply. We shall deal with measurement of voltage first. There are two main points to remember: the method of connecting the voltmeter, and the polarity. A voltmeter must always be connected with the probes *across* the component under test; that is, the circuit need never be broken (see Figure 3-1). This is often referred to as a parallel connection. Knowledge of the polarity of a voltage is essential when using an instrument like the analog multimeter. The red probe must always be connected to a positive terminal, while the black probe is connected to the negative terminal or ground. If the connection is reversed, the analog multimeter will not be able to measure voltage or current. Digital multimeter must likewise follow the same connection. However, when inserted backwards, it will indicate a negative voltage on the readout.

In the measuring of current, the meter must always be inserted into the circuit in such a manner that the current to be measured will flow through the meter (this is often referred to as a series connection.) You must break the circuit to perform a current measurement (see Figure 3-2). If you have not broken the circuit to insert the ammeter, then it is probably not in place correctly. Once correct polarity: the most positive point should be connected to the red terminal when the function switch is set to +dc.

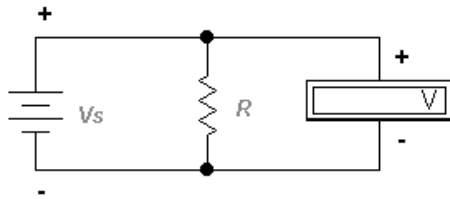


Figure 3-1

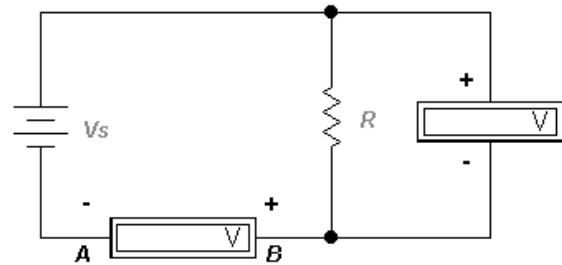


Figure 3-2

PROCEDURES:

1. Switch on the power supply and adjust for minimum output.
2. Set the VOM to measure dc voltage.
3. Connect the VOM directly to the power supply terminals. (Make sure the polarity is correct.)
4. Observe the effect of turning the output voltage control.
5. Adjust the voltage to a value of 1 V.
6. Remove the meter and connect the resistor across the terminals of the power supply as shown in Figure 3-1. Reconnect the meter as shown. (The voltage should not have changed from the 1 V value.)
7. Now break the circuit as shown in Figure 3-2 and insert the other VOM set on the 10 mA current range. (Once again insure that polarity is correct before connecting.)
8. The meter will now be reading the current flowing in the circuit. If the deflection is too small, switch to a smaller range - e.g. 1 mA - to obtain a better reading. (What is the smallest range you may safely use without an over-reading?)
9. Record the current $I(\text{mA})$ in Table 3-1.
10. Increase the voltage in 1 V steps, and for each value of voltage, measure and record the current. Take care to change the range when necessary to avoid damage to the meter movements. (This warning applies only to meters with mechanical movements.)
11. Break the circuit at point A or B and note the effects on both meter readings. (Why does this happen?)

V_s (V)	1	2	3	4	5	6	7	8	9	10
I (mA)										

TABLE 3-1

QUESTIONS:

9. Answer to question in step # 8.

10. Answer to question in step # 11.

11. List the advantages of a digital multimeter over an analog multimeter in measuring dc voltage and current.

CONCLUSIONS

OHM'S LAW
Experiment # 4

Introduction

The experiment on "Ohm's Law" is fundamental to the field of electrical engineering and physics, offering foundational insights into the behavior of electrical circuits. Ohm's Law, a key principle in the study of electricity, states that the current flowing through a conductor between two points is directly proportional to the voltage across the two points, provided the temperature remains constant. This relationship is often represented by the equation $I=V/R$, where I is the current, V is the voltage, and R is the resistance.

This experiment typically involves constructing simple circuits to measure current, voltage, and resistance, thereby allowing students to observe and confirm the linear relationship described by Ohm's Law. Through this practical application, students gain a deeper understanding of the concepts of voltage, current, resistance, and how they interrelate in an electrical circuit.

Learning Objectives

1. Understanding the Fundamentals of Ohm's Law: Develop a thorough understanding of Ohm's Law, including the relationship between voltage, current, and resistance in an electrical circuit.
2. Practical Application of Ohm's Law: Gain hands-on experience by setting up circuits, measuring voltage across and current through resistive elements, and calculating their resistance.
3. Accuracy in Measurements: Learn to accurately measure electrical quantities using appropriate tools like voltmeters, ammeters, and ohmmeters, and understand the significance of precision in these measurements.

4. Interpreting and Analyzing Data: Enhance skills in interpreting the data obtained from experiments, analyzing it in the context of Ohm's Law, and understanding the linear relationship between voltage and current.
5. Circuit Assembly Skills: Develop the ability to correctly assemble simple circuits and understand the function of each component within these circuits.
6. Understanding the Limitations of Ohm's Law: Recognize the conditions under which Ohm's Law is applicable and understand its limitations, especially in non-ohmic materials where the relationship between voltage and current is not linear.
7. Safety in Electrical Experiments: Emphasize the importance of safety while conducting electrical experiments, including the correct and safe use of electrical measuring instruments.
8. Application in Complex Circuits: Apply the understanding of Ohm's Law to more complex circuits, predicting and verifying the behavior of these circuits.
9. Problem-Solving and Critical Thinking: Develop problem-solving skills and critical thinking by applying Ohm's Law to various theoretical and practical scenarios.

By mastering these objectives, students will not only learn the fundamental principle of Ohm's Law but also lay a strong foundation for more advanced studies in electrical circuits and systems.

MATERIALS REQUIRED:

Power Supply: dc power supply 0-10 V

Instruments 2 VOMs (or DVMs)

Resistors: 1 k Ω , 2 k Ω $\pm 5\%$

breadboard

connecting wires

Ohm's Law is the basis of many electrical circuit calculations and is one of the most important theories you will learn: $V = IR$. The purpose of this experiment is to verify Ohm's Law, which, in words, simply says that the current through a resistor is proportional to the voltage across it. The way in which we accomplish this is to measure the voltage across and current through a known resistor for several different pairs of values. We can then plot the data on a graph, and if the relationship is truly linear, it should yield a straight line.

For the graphing part of the experiment, there is a clear choice: we can plot I versus V or V versus I . Both are equally valid, but V versus I gives the value (R) directly as the slope of the line, whereas I versus V gives the reciprocal of R (G). Graph paper is provided so that you can plot either or both for the two resistors in this experiment.

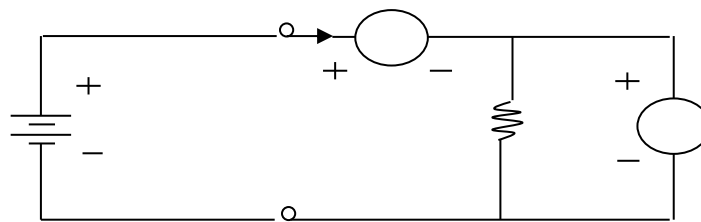


Figure 4 -1

PROCEDURE

1. Connect the circuit in Figure 4 -1 with $R = 1\text{ k}\Omega$.
2. Beginning at 0 V, increase the voltage across R in 1 V steps, and for each case, measure and record the resulting current in Table 4 -1.

3. Use an ohmmeter to measure the actual resistance of R and record this value in Table 4-1 under R_{meas} .
4. Repeat steps 1 through 3 with the 2 k Ω resistor in place. Record the results in Table 4 -2.
5. Plot, on the same scales and axis, graphs of I versus V for each data table. (Assign I to the vertical axis and V to the horizontal axis.)
6. Construct a right triangle on each graph and from this, determine the slope and hence evaluate the conductance G .
7. From this information, evaluate the resistance R . Record G and R from each graph in the appropriate column in Table 4-3.
8. Compare these experimentally obtained values with those measured values recorded in the respective tables.

TABLE 4 -1

Nominal Resistance $R = 1 \text{ k}\Omega$	$V_s \text{ (V)}$	0	1	2	3	4	5	6	7	8	9
$R_{\text{meas.}}$	$I_m \text{ (mA)}$										

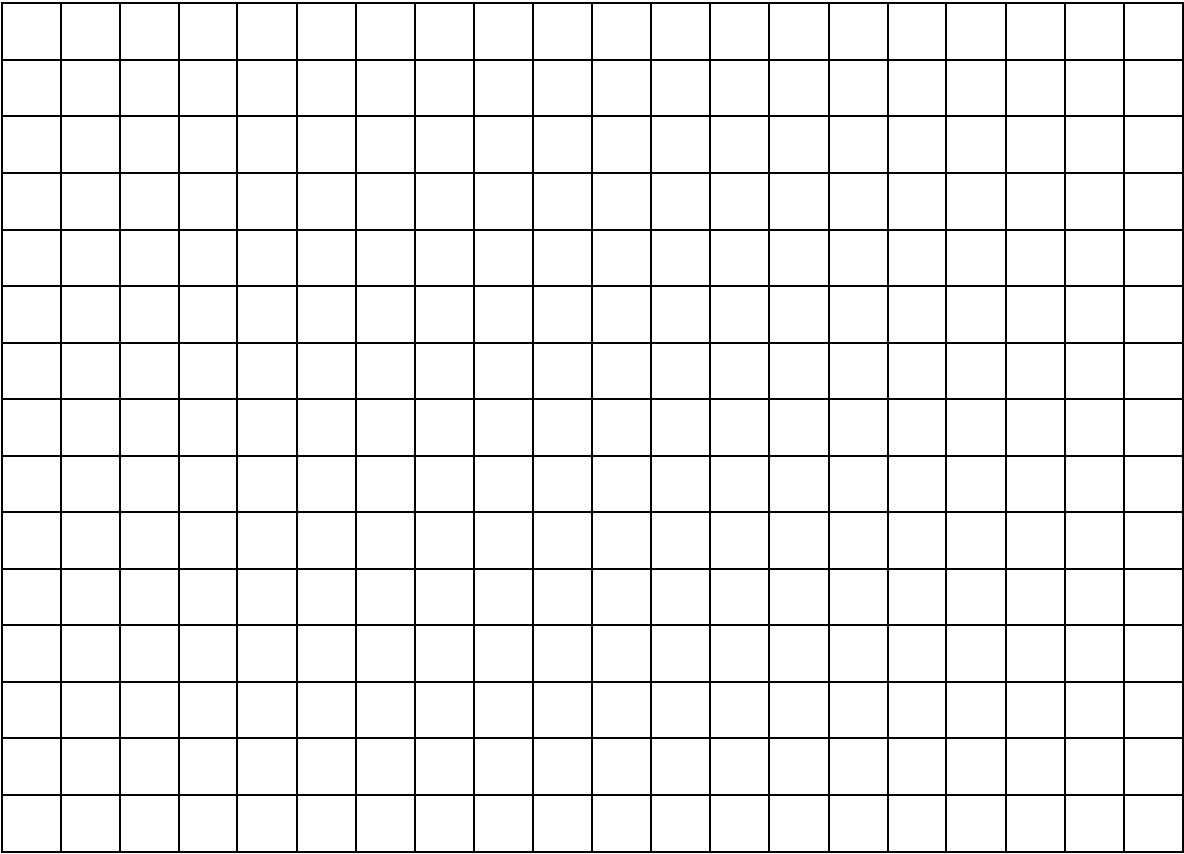
Nominal Resistance $R = 2 \text{ k}\Omega$	$V_s \text{ (V)}$	0	1	2	3	4	5	6	7	8	9
$R_{\text{meas.}}$	$I_m \text{ (mA)}$										

TABLE 4 -2

	Slope (G)	R (1/G)
Table 4-1		
Table 4-2		

TABLE 4 -3

CONCLUSIONS :



POWER IN DC CIRCUITS

Experiment # 5

Introduction

The experiment on "Power in DC Circuits" is a crucial component of education in electrical engineering and physics, focusing on the concept of power in direct current (DC) electrical circuits. Power in electrical terms is defined as the rate at which electrical energy is transferred by an electric circuit. In DC circuits, where the current flows in one direction, the power can be calculated using the formula $P=VI$, where P is power in watts, V is voltage in volts, and I is current in amperes. This fundamental concept is vital for understanding how much energy is being consumed or dissipated in a circuit component, such as a resistor.

The experiment typically involves constructing DC circuits with various components, measuring voltage and current, and subsequently calculating the power. This practical approach helps students understand how different components in a circuit contribute to the overall power usage and how changes in voltage or current affect the power dissipated.

Learning Objectives

1. Understanding the Concept of Power: Develop a solid understanding of electrical power in the context of DC circuits, including how it is calculated and its units of measurement.
2. Practical Measurement of Voltage and Current: Gain hands-on experience in measuring voltage and current in DC circuits using appropriate instruments like voltmeters and ammeters.
3. Calculating Power in Circuits: Learn to calculate the power dissipated or consumed in various components of a DC circuit using the formula $P=VI$ and understand the practical implications of these calculations.

4. Circuit Construction and Analysis: Enhance skills in constructing simple DC circuits and analyzing them to understand how power is distributed and dissipated across different components.
5. Understanding the Role of Resistance in Power Dissipation: Explore the relationship between resistance, voltage, current, and power, and how changing one of these parameters affects the others.
6. Safety in Electrical Experiments: Emphasize safe practices while handling electrical circuits, especially when dealing with higher voltages or currents that can increase power dissipation.
7. Interpreting and Recording Data: Develop proficiency in interpreting data from experiments, recording it accurately, and using it to analyze the circuit's power dynamics.
8. Application of Theoretical Principles: Apply theoretical principles of power to real-world scenarios, understanding the practical applications and implications in various electrical devices and systems.
9. Energy Efficiency Considerations: Understand the importance of energy efficiency in electrical circuits, and how power measurement is crucial in designing energy-efficient systems.

By achieving these objectives, students will gain a comprehensive understanding of how power functions in DC circuits, a knowledge essential for anyone involved in the design, analysis, or application of electrical systems.

MATERIALS REQUIRED:

Power Supply: dc power supply 0-10 V
Instruments 2 analog multimeter (or DMMs)
Resistors: 1 k Ω , 2 k Ω $\pm$ 5 percent ($\frac{1}{4}$ W)

breadboard
connecting wires

Electronic devices and circuits require energy to operate. Power is a measure (in watts) of the energy (in Joules) consumed by a given device in one second. For a resistor, three equations will yield the power dissipated: $P = IV$, $P = V^2 / R$, and $P = I^2 / R$. In this experiment, you will verify these formulae and plot a graph of P versus I to demonstrate the fact that it is a curve. The curve is actually a parabola, and the equation is called a *quadratic*. By plotting two curves on the same scales and axes, you will be able to see which resistance consumes more power for a given current (or voltage).

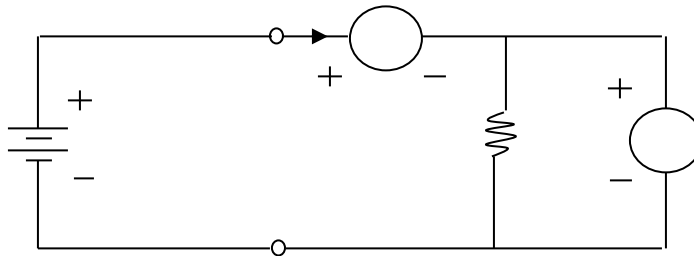


Figure 5 - 1

PROCEDURE:

1. Use an ohmmeter to measure the resistance of the 1 k Ω resistor. Record this measured value in Table 5 -1.
2. Connect the circuit in Figure 5 -1 and measure the current for each value of V_s in Table 5 -1. Record these measured currents in Table 5 -1.

3. Use the voltage and current data together with the measured value of R to complete the rest of Table 5 -1 for the power P . All power quantities should agree.
4. Repeat steps 1 through 3 with the 2 k Ω resistor, and record all data in Table 5 -2.
5. Plot graphs as follows: On the same scales and axes, plot P versus I for both the 1 k Ω and 2 k Ω resistors.

V_s (V)	0	1	2	3	4	5	6	7	8	9	10
I (mA)											
$P (= IV)$											
$P (= I^2 R)$											
$P (= V^2/R)$											
	$R_{\text{meas}} =$										

TABLE 5 -1

V_s (V)	0	1	2	3	4	5	6	7	8	9	10
I (mA)											
$P (= IV)$											
$P (= I^2 R)$											
$P (= V^2/R)$											
	$R_{\text{meas}} =$										

TABLE 5-2

QUESTIONS:

1. In a circuit such as that in Figure 6-1, if the voltage is maintained constant and R decreases, the power in R
 - (a) Decrease
 - (b) Increases
 - (c) Stays the same

2. If the voltage across a resistance is doubled, the power increases by a factor of
(a) 2 (b) 4 (c) 1 (d) None of these
3. For a constant voltage, power is directly proportional to resistance.
(a) True (b) False
4. For a P versus I curve as was plotted in this experiment, the slope is equal to the
(a) Resistance (b) Conductance (c) Neither a nor b
5. Show that if two unequally valued resistors have the same power rating, the smaller-valued of the two will have a greater maximum safe current value.

6. When a resistor is connected to the terminals of a regulated power supply as in this experiment, you are more likely to exceed a small-valued resistor's power rating than a larger-valued one of the same rating. Explain.

CONCLUSION:

SERIES RESISTIVE CIRCUITS
Experiment # 6

Introduction

The experiment on "Series Resistive Circuits" is a foundational study in the field of electrical engineering and electronics, focusing on the behavior and characteristics of resistors connected in a series arrangement. In a series circuit, resistors are connected end-to-end, so the same current flows through each resistor, but the voltage across each resistor can be different. This type of circuit arrangement is fundamental and widely used in various electronic applications. Understanding how resistors behave in series is crucial for analyzing and designing more complex circuits.

This experiment typically involves constructing circuits with resistors connected in series, measuring the voltage across each resistor, and the total current flowing through the circuit. Students learn to apply Ohm's Law and the principles of series circuits to analyze these measurements. The experiment enhances understanding of concepts like voltage division, total resistance in a series circuit, and the effect of adding resistors in series.

Learning Objectives

1. Understand the Concept of Series Circuits: Gain a comprehensive understanding of what constitutes a series circuit and the fundamental characteristics of resistive components connected in series.
2. Circuit Construction and Analysis: Develop the ability to construct series resistive circuits and analyze them, including calculating the total resistance and understanding how current flows in the circuit.
3. Application of Ohm's Law: Apply Ohm's Law ($V=IR$) to individual resistors and the entire series circuit to determine voltages, currents, and resistances.

4. Voltage Division Principle: Understand and apply the principle of voltage division in series circuits, which explains how the total voltage is divided among the resistors.
5. Measurement of Electrical Quantities: Learn to accurately measure voltage across individual resistors and the total current in the circuit using voltmeters and ammeters.
6. Interpreting and Recording Data: Enhance skills in interpreting data obtained from experimental measurements, recording it accurately, and using it for circuit analysis.
7. Calculating and Understanding Total Resistance: Develop the ability to calculate the total resistance in a series circuit and understand how adding or removing a resistor affects the overall resistance.
8. Safety in Circuit Experiments: Emphasize safe practices in handling and constructing electrical circuits to prevent accidents and equipment damage.
9. Real-World Applications: Understand the practical applications of series resistive circuits in real-world electronic devices and systems.

By mastering these objectives, students will have a thorough understanding of the principles and practical applications of series resistive circuits, a fundamental concept in the field of electrical and electronic engineering.

MATERIALS REQUIRED:

Power Supply: dc power supply 0-10 V

Instruments: 1 VOM or DVM

Resistors: (1/4 W) 1 k Ω , 3.9 k Ω , 2 k Ω , 5.1 k Ω , 3 k Ω $\pm 5\%$

When resistors are connected in series, the current that will flow is calculated from a knowledge of the total resistance. The total resistance in this case is simply the sum of the

individual resistors. By knowing the total resistance, the voltage required for a desired current or the current resulting from an applied voltage can easily be calculated.

One of the purposes of this experiment is to provide you with still more experience in using your multimeter. You will use it on the ohms, volts, and milliamps functions in verifying the Series Resistor Rule.

PROCEDURE:

1. Calculate the upper and lower limits of the resistance for each resistor and record these in Table 6 - 1.
2. Use the ohmmeter to measure the actual values of each resistor and record these under R_{meas} .
3. Complete Table 6 -1 by calculating the upper and lower limits to $R_T (= R_1 + R_2 + R_3 + R_4 + R_5)$, and then measuring the actual value of R_T by series connecting the resistors. Does the measured value equal the sum of the individual measured values? If not, why not?
4. Using a value for V_S of 10 V, calculate the nominal, minimum, and maximum values for I in the circuit of Figure 6 -1 and record them in Table 6 -2.
5. Connect the circuit in Figure 6 -1 and adjust V_S to a value of 10 V. Measure the current I and record it under I_{meas} in Table 6 -2. (Does it appear consistent with R_{meas} ?)
6. Using a value for I of 0.5 mA, calculate the nominal, minimum, and maximum values for V_S in order to sustain this current through the series combination. Record these in Table 6 -3.
7. Connect the circuit and adjust V_S until $I = 0.5$ mA. Measure the voltage and record it in Table 6 -3. (Does it appear consistent with R_{meas} ?)

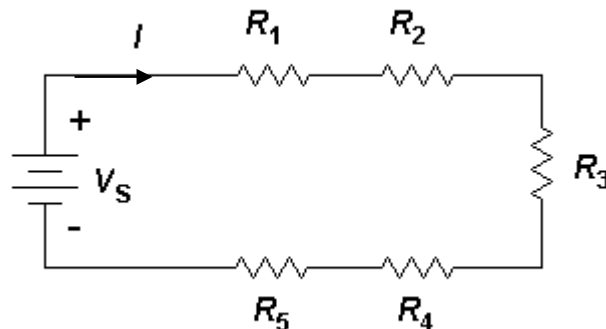


Figure 6 - 1

R	1 K Ω	2 K Ω	3 K Ω	3.9 K Ω	5.1 K Ω	R _T = 15 K Ω
R_{min}						
R_{max}						
R_{meas}						

TABLE 6 - 1

$V_S = 10$ V	I_{nom}	
	I_{min}	
	I_{max}	
	I_{meas}	

TABLE 6 - 2

$I_S = 10$ A	V_{nom}	
	V_{min}	
	V_{max}	
	V_{meas}	

TABLE 6 - 3

QUESTIONS:

1. If the measured value of R_T did not equal the sum of the individually measured resistances, you might conclude that
 - (a) The resistors had changed in value
 - (b) The resistors were out of tolerance
 - (c) The ohmmeter measurements are subject to error
 - (d) All of the above
2. Three resistors of value 1 k Ω , 2 k Ω , and 3 k Ω , each having a tolerance of ± 5 percent, are series connected across a 10 V source. The smallest current you would expect to flow in the circuit is
 - (a) 1.59 mA
 - (b) 1.67 mA
 - (c) 1.75 mA
 - (d) None of these
3. The current through each element in a series-connected circuit such as that in Figure 8-1 is the same.
 - (a) True
 - (b) False
4. If the resistor tolerances in this experiment had been ± 10 percent, the maximum current through the series combination with a 10 V source would be been
 - (a) 61 mA
 - (b) 606 μ A
 - (c) 741 μ A
 - (d) 667 mA
5. If any resistor in a series circuit is reduced in value, the total current will increase. Explain why this is so.

6. If two resistors in a series circuit suffer a change in resistance such that one increases by 10 and the other decreases by 10 percent, explain why the current will not necessarily stay constant. Under what conditions would the current undergo no change?

CONCLUSION:

PARALLEL RESISTIVE CIRCUITS
Experiment # 7

Introduction

The experiment on "Parallel Resistive Circuits" is an essential part of learning in electrical engineering and electronics, focusing on the behavior and characteristics of resistors connected in parallel. In a parallel circuit configuration, resistors are connected in such a way that each resistor is placed across the same voltage source, thus each resistor has the same voltage across it, but the current flowing through each resistor can be different. This configuration is crucial in various electronic applications due to its unique properties, such as maintaining a uniform voltage across different circuit components.

Conducting experiments with parallel resistive circuits involves setting up circuits with resistors connected in parallel, measuring individual and total currents, and the voltage across each resistor. This practical approach helps students understand concepts like current division, total resistance in parallel circuits, and the effect of adding resistors in parallel on the overall circuit behavior.

Learning Objectives

1. Understand the Concept of Parallel Circuits: Gain a comprehensive understanding of parallel circuits, including the fundamental characteristics of resistive components connected in parallel.
2. Circuit Construction and Analysis: Develop skills in constructing parallel resistive circuits and analyzing them, focusing on calculating the total resistance and understanding how voltage is distributed in the circuit.

3. Application of Ohm's Law and Kirchhoff's Current Law: Apply Ohm's Law ($V=IR$) and Kirchhoff's Current Law to understand how current is divided among the resistors and calculate voltages and currents in the circuit.
4. Current Division Principle: Understand and apply the principle of current division in parallel circuits, which explains how the total current is divided among the resistors.
5. Measurement of Electrical Quantities: Learn to accurately measure current through individual resistors and the total voltage across the circuit using ammeters and voltmeters.
6. Interpreting and Recording Data: Enhance the ability to interpret data obtained from experimental measurements, record it accurately, and use it for circuit analysis.
7. Calculating and Understanding Total Resistance: Develop the ability to calculate the total resistance in a parallel circuit and understand how adding or removing a resistor affects the overall resistance.
8. Safety in Circuit Experiments: Emphasize safe practices in handling and constructing electrical circuits to prevent accidents and damage to equipment.
9. Real-World Applications: Understand the practical applications of parallel resistive circuits in real-world electronic devices and systems, such as in power distribution and load balancing.
10. By mastering these objectives, students will acquire a thorough understanding of the principles and practical applications of parallel resistive circuits, which is fundamental for the design and analysis of more complex electrical and electronic systems.
11. To verify that the total resistance in a parallel-connected circuit is given by the reciprocal rule.

MATERIALS REQUIRED:

Power Supply: dc power supply 0-10 V

Instrument: 1 VOM or DVM

Resistors: (1/4 W), 1.2 k Ω , 2.4 k Ω , 1.5 k Ω , 3 k Ω , 2 k Ω , $\pm 5\%$

In a parallel circuit, each component has the same voltage across it, and the currents through each component are independent of one another. The standard procedure of analyzing a parallel circuit is to apply Ohm's Law to each branch and obtain the currents. Alternatively, the total current can be obtained directly using the total resistance concept. When there are only two resistors, the total resistance is most easily calculated by the Product-Over-Sum rule.

In this experiment, you will predict the total current that should flow into a simple two-resistor parallel circuit, given the source voltage. Then you will work in reverse and calculate the source voltage required for a given total current.

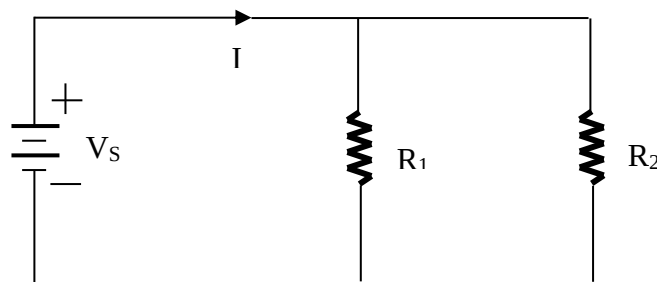


Figure 7 -1

PROCEDURE

1. Calculate the upper and lower limits of the resistance for each resistor and record these in Table 7 -1.
2. Use the ohmmeter to measure the actual values of each resistor and record these under R_{meas} .
3. Complete Table 7 -1 by calculating the nominal, minimum, and maximum values for the 3 parallel combinations, and then measuring the resistance by connecting these combinations.
4. Using a value for V_s of 5 V, calculate the nominal, minimum, and maximum values for I in the circuit of Figure 7 -1 and record them in Table 7 -2 for each parallel combination.

5. Connect the circuit and adjust V_S to a value of 5 V. Measure the current I and record it under I_{meas} in Table 7 -2. Do this for each parallel combination. (Does it appear consistent with R_{meas} ?)

6. Using a value for I of 5 mA, calculate the nominal, minimum, and maximum values for V_S in order to sustain this current through the parallel combination. Record these in Table 7 -3.

7. Connect the circuit and adjust V_S until $I = 5$ mA. Measure the voltage and record it in Table 7 -3. (Does it appear consistent with R_{meas} ?) Do this for each parallel combination.

R_{nom}	R_1	R_2	R_3	R_4	R_5	$R_5 \parallel R_2$	$R_4 \parallel R_1$	$R_5 \parallel R_3$
	1.2 K Ω	1.5 K Ω	2 K Ω	2.4 K Ω	3K Ω			
R_{min}								
R_{max}								
R_{meas}								

TABLE 7 -2

$V_S = 5$ V		$R_5 \parallel R_2$	$R_4 \parallel R_1$	$R_5 \parallel R_3$
	I_{nom}			
	I_{min}			
	I_{max}			
	I_{meas}			

TABLE 7 – 1

$I = 5\text{ mA}$		$R_5 \parallel R_2$	$R_4 \parallel R_1$	$R_5 \parallel R_3$
	V_{nom}			
	V_{min}			
	V_{max}			
	V_{meas}			

TABLE 7 - 3

QUESTIONS:

1.If all the resistors given to you in this experiment were placed in parallel, the resulting total resistance would be nominally

- (a) 324 Ω (b) 364 Ω
(c) 1200 Ω (d) 3.6 k Ω

2.The total resistance of a parallel combination of resistors will always be less than the smallest resistor in the combination.

- (a) True (b) False

3.The tolerance of *any* parallel combination of resistors (all with the same tolerance) is the same as the individual resistor tolerance.

- (a) True (b) False

4.Consider a parallel connection of two resistors of values $R\Omega$ and $3 R\Omega$. Both have a 10 percent tolerance. The maximum expected total resistance is

- (a) 4.4 $R\Omega$ (b) 0.73 $R\Omega$
(c) 0.83 $R\Omega$ (d) 0.75 $R\Omega$

5.Explain why the addition of a parallel resistance to any circuit will always make the total current increase.

CONCLUSION:

SERIES-PARALLEL COMBINATIONS
Experiment # 8

Introduction

The experiment on "Series-Parallel Combinations" is a critical educational component in electrical engineering and electronics, blending the concepts of series and parallel circuits into more complex configurations. Series-parallel circuits are a ubiquitous aspect of real-world electronic devices, where components are often arranged in a combination of both series and parallel connections to achieve desired electrical characteristics. Understanding these circuits is essential for designing and troubleshooting a wide range of electronic systems.

In a series-parallel circuit, some components are connected in series (where current flows through each component sequentially), while others are connected in parallel (where components share the same voltage). This arrangement can create circuits with varying resistance, voltage, and current characteristics, which are not possible with purely series or purely parallel circuits. The experiment typically involves constructing and analyzing circuits that incorporate both series and parallel elements, providing practical insights into how these combined circuits behave.

Learning Objectives

1. Understanding Series-Parallel Circuit Concepts: Develop a thorough understanding of the characteristics of series-parallel circuits, including how components in these circuits share voltage and current.
2. Circuit Construction and Analysis: Gain practical experience in constructing series-parallel circuits and analyzing them to understand the overall behavior of the circuit, including voltage distribution, current flow, and total resistance.

3. Application of Ohm's Law and Kirchhoff's Rules: Apply Ohm's Law and Kirchhoff's Voltage and Current Laws to calculate voltages, currents, and resistances in different parts of a series-parallel circuit.
4. Measurement of Electrical Quantities: Learn to accurately measure voltage, current, and resistance in series-parallel circuits using appropriate instruments like voltmeters, ammeters, and ohmmeters.
5. Interpreting and Recording Data: Enhance skills in interpreting data obtained from experimental measurements, recording it accurately, and using it for circuit analysis and troubleshooting.
6. Calculating Total Resistance: Develop the ability to calculate the total resistance in series-parallel circuits, understanding how different configurations affect the overall resistance.
7. Problem-Solving and Critical Thinking: Foster problem-solving and critical thinking skills by analyzing complex circuit behaviors and determining the impact of changing circuit components or configurations.
8. Safety in Circuit Experiments: Emphasize the importance of safety practices in handling and constructing electrical circuits.
9. Real-World Applications: Recognize the practical applications and significance of series-parallel circuits in various electronic devices and systems, enhancing the understanding of how theoretical concepts are applied in practical scenarios.

By mastering these objectives, students will not only comprehend the complex behaviors of series-parallel circuits but will also be equipped with the skills necessary for designing, analyzing, and troubleshooting sophisticated electronic circuits and systems.

MATERIALS REQUIRED:

Power Supply: dc power supply 0-10 V

Instrument: 1 VOM or DVM

Resistors: (1/4 W) 2 k Ω , 12 k Ω , 24 k Ω $\pm 5\%$

The most general kind of circuit connection is that which contains both series and parallel resistor configurations. The standard method of attack in this kind of problem is to first combine parallel resistors into a single equivalent, then add the series resistors.

The circuit in this experiment (see Figure 8 -1) contains a simple combination of two parallel resistors in series with a third resistor. Using Kirchhoff's Voltage Law, Kirchhoff's Current Law, and Ohm's Law, you can solve for all currents and voltages. In this case, you will calculate theoretical values for all of the quantities and verify them experimentally.

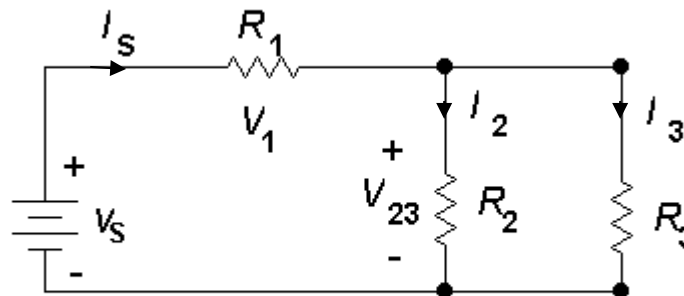


Figure 8 - 1

PROCEDURE

- Calculate the upper and lower limits of the resistance for each resistor and record these in Table 8 -1.
- Use the ohmmeter to measure the actual values of each resistor and record these under R_{meas} .
- Complete Table 8 -1 by calculating the nominal, minimum, and maximum values for the total circuit resistance R_T of the circuit in Figure 8 -1, and then measuring this resistance using an ohmmeter.
- Using a value of V_S of 10 V, calculate the nominal, minimum, and maximum values for I_S in the circuit of Figure 8 -1 and record them in Table 8 -2.

- e. Connect the circuit and adjust V_S to a value of 10 V. Measure the current I_S , and record it under I_{meas} in Table 8 -2. (Does it appear consistent with R_{meas} ?)
- f. Measure the remaining currents and voltages and record them in Table 8 -2.
- g. Using a value for I_S of 0.5 mA, calculate the nominal, minimum, and maximum values for V_S in order to sustain this current through the series-parallel combination. Record these values in Table 8 - 2.
- h. Connect the circuit and adjust V_S until $I = 0.5$ mA. Measure the voltage and record it in Table 8 -3. (Does it appear consistent with R_{Tmeas} ?)
- i. Measure the remaining currents and voltages and record them in Table 8 - 2.

D. R	$R_1 = 2 \text{ k}\Omega$	$R_2 = 12 \text{ k}\Omega$	$R_3 = 24 \text{ k}\Omega$	$R_T = R_1 + R_2 \parallel R_3$
R_{min}				
R_{max}				
R_{meas}				

TABLE 8 -1

$V_S = 10\text{ V}$	I_{nom}		$I = 0.5\text{ mA}$	$V_{S\text{ nom}}$	
	I_{min}			$V_{S\text{ min}}$	
	I_{max}			$V_{S\text{ max}}$	
	I_{meas}			$V_{S\text{ meas}}$	
	$I_{2\text{ meas}}$			$I_{2\text{ meas}}$	
	$I_{3\text{ meas}}$			$I_{3\text{ meas}}$	
	$V_{1\text{ meas}}$			$V_{1\text{ meas}}$	
	$V_{23\text{ meas}}$			$V_{23\text{ meas}}$	

TABLE 8 - 2

QUESTIONS:

- In Figure 8 -1, if R_2 were less than nominal, then for a given voltage V_S ,
 - I_2 would be less than nominal
 - I_3 would be less than nominal
 - I_5 would be less than nominal
 - I_3 would be greater than nominal
- If an additional resistor R_4 were added in parallel with R_1 , then the total resistance R_T would
 - Decrease
 - Increase
 - Not change
- In this circuit, if the value of R_3 were doubled, then for a given voltage V_S ,
 - I_3 would halve
 - I_S and I_3 would halve
 - I_2 would be unaffected
 - All currents would be affected
- If all of the resistors in this circuit have a tolerance of ± 5 percent, then the tolerance of R_T will be
 - $\pm 5\%$
 - $\pm 7.5\%$
 - $\pm 15\%$
 - None of these

5. Decreasing any resistor value in such a circuit will always decrease the total resistance R_T . Is this true? Explain?

6. How many different possible values of R_T are obtainable with the three resistors if each circuit is to be a series-parallel combination of the kind in Figure 8 -1? Calculate all of the possible values using the resistors in this experiment.

CONCLUSIONS:

KIRCHHOFF'S VOLTAGE LAW
Experiment # 9

Introduction

The experiment on "Kirchhoff's Voltage Law" (KVL) is a pivotal educational exercise in the fields of electrical engineering and electronics. Kirchhoff's Voltage Law, a fundamental principle in circuit theory, states that the sum of the electrical voltages around any closed network (or loop) is zero. This law is crucial for analyzing and understanding the behavior of electrical circuits. It arises from the conservation of energy principle, which in electrical terms implies that the total energy gained and lost as electric current travels around a loop must be equal.

This experiment involves constructing various electrical circuits and applying KVL to determine the voltages across different components. By doing so, students gain a deeper understanding of how voltage is distributed in a circuit and how to apply this law for circuit analysis. This practical approach not only reinforces theoretical knowledge but also enhances skills in circuit construction and problem-solving.

Learning Objectives

1. Understanding Kirchhoff's Voltage Law: Develop a comprehensive understanding of KVL, including its theoretical basis and applications in circuit analysis.
2. Circuit Construction for KVL Application: Gain experience in constructing electrical circuits that are suitable for the application of KVL, including simple and more complex loops.
3. Measurement of Voltage in Circuits: Learn to accurately measure voltage across various components in a circuit, using voltmeters and understanding the significance of polarity in voltage measurements.
4. Application of KVL in Circuit Analysis: Apply KVL to analyze circuits, calculate unknown voltages, and verify the law in practical scenarios.
5. Problem-Solving and Critical Thinking: Enhance problem-solving and critical-thinking skills by interpreting circuit diagrams, identifying loops, and applying KVL to solve for unknown values.
6. Understanding the Role of Components in Voltage Drops: Comprehend how different components like resistors, capacitors, and inductors contribute to the voltage drops in a circuit.
7. Safety in Handling Electrical Equipment: Emphasize the importance of safety practices in constructing and analyzing electrical circuits.

8. Data Analysis and Interpretation: Develop skills in recording experimental data, analyzing it, and comparing it with theoretical calculations.
9. Real-World Applications of KVL: Understand the practical applications of KVL in real-world scenarios, including its importance in designing and troubleshooting electrical circuits and systems.

By mastering these objectives, students will not only grasp the essentials of Kirchhoff's Voltage Law but will also acquire practical skills and knowledge crucial for analyzing complex electrical circuits and systems in various engineering fields.

EQUIPMENT: dc power supply 0-10 V
1 VOM or DVM
Resistors (1/4 W): 1.2 k Ω 4.7 k Ω
 1.8 k Ω 5.6 k Ω
 2.4 k Ω 6.8 k Ω
 3.3 k Ω 8.2 k Ω
 ± 5 percent

BACKGROUND: Kirchhoff's Voltage Law states that the algebraic sum of voltages around a closed path is equal to zero. With regard to Figure 10-1, this means that

$$V_S - V_1 - V_2 - V_3 - V_4 = 0$$

Or

$$V_S = V_1 + V_2 + V_3 + V_4$$

By connecting such a circuit and measuring the voltages, it should be possible to verify this relationship. In performing an experiment of this nature, you should remember that each measurement is subject to error, and when you sum such measurement, the errors may add and produce what may appear to be an inconsistency. The important thing to remember is that, if the instruments were perfect, we should obtain a sum of voltage drops *exactly* equal to the source voltage.

PROCEDURE:

- j. Select four resistors from the eight available, and connect them in series as shown in Figure 10-1. Record your selected values in Table 10-1.
- k. Using the total resistance R_T and resulting current I , calculate and record in Table 10-1 the voltages V_1 , V_2 , V_3 , and V_4 for a source voltage of 10 V.
- l. Connect your circuit and adjust the source voltage to a value of 10 V.
- m. Measure the voltage across each resistor, and record the values in the appropriate columns of Table 10-1. Indicate also the magnitudes and polarities of the voltages in the figure that accompanies the table.
- n. Record the sum of these voltages under "sum" in the table. (Is the sum consistent with the voltage V_S ?)

- o. Measure and record the current I in the circuit.
- p. Repeat steps
different
record all

1 through 6 with a
resistor selection and
results in table 10-2.

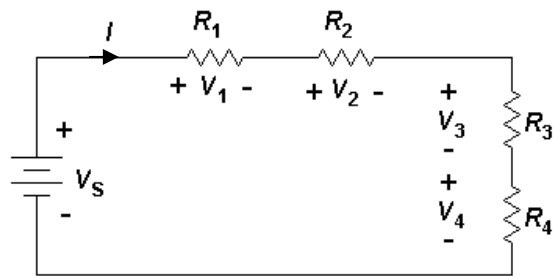


FIGURE 10-1

TABLE 10-1

	R_1	R_2	R_3	R_4	R_T	Measured Current I (mA)
<i>Selected Nominal Values</i>						
Calculated Voltage Values	V_1	V_2	V_3	V_4		
Measured Voltage	V_1	V_2	V_3	V_4	V_{SUM}	

TABLE 10-2

	R_1	R_2	R_3	R_4	R_T	Measured Current I (mA)
<i>Selected Nominal Values</i>						
Calculated Voltage Values	V_1	V_2	V_3	V_4		

14. Is it possible to have some voltages less than nominal and some voltages greater than nominal simultaneously in such a circuit? If so, how?

CONCLUSIONS:

VOLTAGE DIVIDERS
Experiment # 10

Introduction

The experiment on "Voltage Dividers" is a critical and foundational study in electrical engineering and electronics, focusing on a circuit configuration used to obtain a voltage lower than the input voltage. A voltage divider is a simple circuit made up of resistors in series through which the input voltage is applied. The output voltage is taken across one of the resistors. This basic principle is widely used in various applications, such as adjusting signal levels, setting reference voltages, and scaling down voltages for measurement purposes.

Understanding how voltage dividers work and how to design them is crucial for any student or professional in the field of electronics. The experiment typically involves constructing voltage divider circuits, calculating expected output voltages based on resistor values, and then measuring the actual output voltages to compare with theoretical predictions. This hands-on approach helps in understanding the practical aspects and limitations of voltage dividers.

Learning Objectives

1. Understand the Principle of Voltage Dividers: Develop a thorough understanding of the voltage divider rule, its basic configuration, and how it functions in a circuit.
2. Circuit Construction and Analysis: Learn to construct voltage divider circuits using resistors and analyze them to predict the output voltage based on given input voltages and resistor values.
3. Application of Ohm's Law and Series Circuit Principles: Apply Ohm's Law and series circuit principles to calculate the voltage drops across individual resistors and determine the output voltage.
4. Measurement and Comparison of Voltages: Gain hands-on experience in measuring voltages in a circuit using voltmeters and comparing the measured values with theoretical calculations.
5. Understanding the Impact of Load on Voltage Dividers: Study how connecting a load to the output of a voltage divider affects the output voltage and learn to account for this in circuit design.
6. Analyzing Non-ideal Behaviors: Recognize and analyze the non-ideal behaviors and limitations of voltage dividers, such as the effects of internal resistance and loading.
7. Safety Practices in Circuit Experiments: Emphasize safe practices while handling and measuring electrical circuits.

8. Problem-Solving and Design Skills: Develop problem-solving skills by designing voltage dividers for specific output requirements and troubleshooting issues in the circuit.
9. Real-World Applications: Understand the practical applications of voltage dividers in electronic devices and systems, enhancing the relevance of theoretical concepts.

By mastering these objectives, students will not only gain a solid understanding of the principles of voltage dividers but will also acquire practical skills necessary for designing and analyzing circuits in various electronic applications.

EQUIPMENT: dc power supply 0-10 V
 1 VOM or DVM
 Potentiometer 10 k Ω
 Resistors (1/4 W): 1.6 k Ω 3.3 k Ω
 2.4 k Ω ± 5 percent
 2.7 k Ω

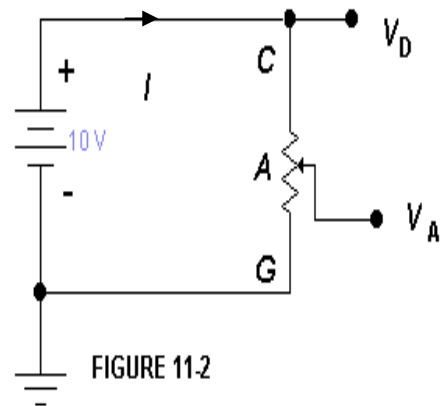
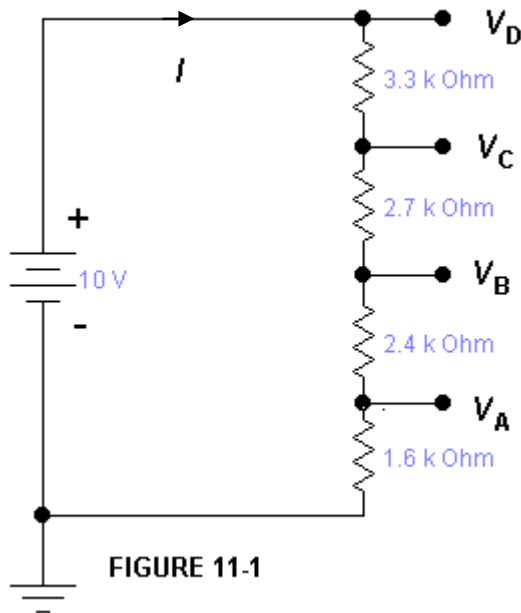
Voltage dividers find many applications in electronic circuits. The requirement for voltages of different values often arises and is normally accomplished by a series resistive circuit as shown in Figure 11-1. Voltages V_A , V_B , V_C , and V_D are potentials measured with respect to ground (the negative side of the battery). Since the current in each element is the same, by Ohm's Law the individual voltage across each resistor is proportional to the respective resistor value.

Figure 11-2 shows how a potentiometer can be used as a continuous voltage divider. For this experiment its value has been chosen to be 10 k Ω , which is exactly the sum of the four resistors in Figure 11-1. By appropriate positioning of the wiper arm, any of the voltages you obtained in the "discrete resistor" circuit of Figure 11-1 (and all values in between) can be derived. Potentiometers such as these are often used in this manner as volume controls in audio equipment.

PROCEDURE

1. For the circuit in Figure 11-1, calculate the nominal values for the voltages V_A , V_B , V_C , and V_D , and record them in Table 11-1. All voltages are with respect to ground.
2. Construct the circuit, and record the measured values of V_A , V_B , V_C , and V_D in Table 11-1 for comparison.*
3. Complete the table by calculating the nominal current I , and then measuring the actual current for comparison.
4. Dismantle the circuit in Figure 11-1 and construct that of Figure 11-2. Place the VOM across the arm and the grounded end of the potentiometer.
5. Measure and record in Table 11-2 the minimum and maximum voltage for V_A as the arm of the potentiometer is moved from one extreme to the other.
6. Adjust the arm of the potentiometer until the voltage V_A is equal to 1.6 V (see first column of Table 11-3). Remove the potentiometer and measure the resistance between points A and G, then A and C. Record these values in Table 11-3.

7. Repeat step 6 for the remaining voltages in Table 11-3.
8. Compare the data obtained in steps 4 through 7 with that obtained in steps 1 through



DATA FOR

TABLE 11-1

Voltage	V_A	V_B	V_C	V_D	Current I	
Nominal					Nominal	
Measured					Measured	

TABLE 11-2

<i>Voltage</i>	$V_A \text{ min}$	$V_A \text{ max}$
<i>Nominal</i>		
<i>Measured</i>		

TABLE 11-3

<i>Voltage</i>	$V^{1.6}$	V^4	$V^{6.7}$	V^{10}
R_{AG}				
R_{AC}				

QUESTIONS:

9. If all the resistance values in Figure 11-1 were stepped up by a factor of 10, the new voltages would
 - i. Be 10 times their old values
 - ii. Not change
 - iii. Be one-tenth their old values
10. When resistor tolerance is taken into account, the smallest expected value for V_C in Figure 11-1 is
 - (a) 6.81 V (b) 6.48 V (c) 6.37 V (d) 6.57 V
11. If the 2.4 k Ω resistor in Figure 11-1 were mistakenly replaced with a 240 Ω resistor, the voltages that would be affected would be
 - (a) V_A and V_B only (b) V_B only
 - (c) V_A , V_B , V_C , and V_D (d) V_A , V_B , and V_C only
12. The current I in Figure 11-2 will change as the arm of the potentiometer is moved.
 - (a) True (b) False
13. Discuss the effects of using high values *versus* low values (with the same ratios) for the resistors in a voltage divider chain.

14. How much power must the potentiometer in Figure 11-2 be capable of handling?
Does the power dissipated depend on the position of the wiper arm?

CONCLUSIONS:

KIRCHHOFF'S CURRENT LAW
Experiment #11

Introduction

The experiment on "Kirchhoff's Current Law" (KCL) is an integral part of electrical engineering and electronics education, focusing on a fundamental principle in the analysis of electrical circuits. Kirchhoff's Current Law states that the total current entering a junction or node in a circuit must equal the total current leaving the node. This law is based on the principle of conservation of charge, implying that charge cannot accumulate at a node within a steady-state circuit. Understanding KCL is essential for analyzing complex circuits, particularly in determining the flow of current in various branches of a network.

This experiment typically involves setting up circuits with multiple branches and nodes, then measuring and calculating the currents in different parts of the circuit. Through this practical application, students gain a deeper comprehension of how current is distributed in a circuit and how KCL can be used for efficient circuit analysis.

Learning Objectives

1. Understanding Kirchhoff's Current Law: Develop a comprehensive grasp of KCL, including its theoretical basis and how it applies to various types of circuits.
2. Circuit Construction for KCL Application: Gain practical experience in constructing circuits with multiple branches and nodes suitable for the application of KCL.
3. Measurement of Current in Circuit Branches: Learn to accurately measure the current in different branches of a circuit using ammeters and understand the significance of direction of current flow.
4. Application of KCL in Circuit Analysis: Apply KCL to analyze and solve for unknown currents in complex circuit configurations.
5. Problem-Solving and Critical Thinking Skills: Enhance problem-solving and critical thinking abilities by identifying appropriate nodes and branches for the application of KCL and determining the direction of current flow.
6. Understanding the Impact of Components on Current Flow: Comprehend how different components like resistors, capacitors, and inductors affect the current distribution in a circuit.
7. Safety in Electrical Experiments: Emphasize the importance of safety practices in constructing and analyzing electrical circuits.

8. Data Interpretation and Analysis: Develop the ability to interpret and analyze data from experiments, and compare it with theoretical calculations and predictions.
9. Real-World Application of KCL: Understand the practical applications of KCL in real-world scenarios, such as in the design and troubleshooting of electrical circuits and systems.

By mastering these objectives, students will not only learn the fundamental principles of Kirchhoff's Current Law but will also be equipped with essential skills for analyzing and understanding complex electrical circuits in various engineering applications.

EQUIPMENT: dc power supply 0-10 V
 1 VOM or DVM
Resistors (1/4 W): 1.2 k Ω 4.7 k Ω
 1.8 k Ω 5.6 k Ω
 2.4 k Ω 6.8 k Ω
 3.3 k Ω 8.2 k Ω
 ± 5 percent

BACKGROUND: Kirchhoff's Current Law states that the algebraic sum of currents into a node is equal to zero. With reference to Figure 11-1 this implies

$$I - I_1 - I_2 - I_3 - I_4 = 0$$

or

$$I = I_1 + I_2 + I_3 + I_4$$

By connecting such a circuit and measuring the currents, it should be possible to verify the above relationship. Once again, as with experiment 10, you should be aware that each individual measurement is subject to error and that the measured currents will probably not sum exactly to the measured total current.

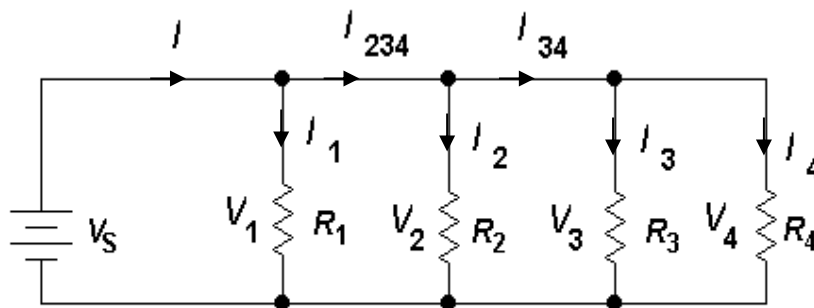


Figure 11-1

PROCEDURE:

1. Select four resistors from the eight available, and connect them in parallel as shown in Figure 11-1. Record your selected values in Table 14-1. Calculate and record R_T .
2. Adjust the source voltage for a value of 10 V.
3. Measure and record the currents I_1 through I_4 in the respective columns of Table 11-1.
4. Measure and record also the currents I_{34} , I_{234} , and I .
5. Calculate the total resistance R_T and the individual resistances R_1 , R_2 , R_3 , and R_4 using the measured values of voltage and current in each case. Compare the results with the nominal values.
6. Calculate the currents I_{34} , I_{234} , and I by adding the respective measured branch currents in each case. Is Kirchhoff's Current Law (KCL) satisfied within the limits of experimental error?
7. Repeat steps 1 through 6 with the remaining four resistors and record all results in Table 11-2.

DATA FOR EXPERIMENT 11

TABLE 11-1

<i>Selected Nominal Values</i>	R_1	R_2	R_3	R_4	R_T		
<i>Measured Currents</i>	I_1	I_2	I_3	I_4	I_{34}	I_{234}	I
<i>Calculated Resistance Values</i>	R_1	R_2	R_3	R_4	R_T		
<i>Calculated Currents by KCL</i>	I_{34}		I_{234}		I		

TABLE 11-2

Selected Nominal Values	R_1	R_2	R_3	R_4	R_T		
Measured Currents	I_1	I_2	I_3	I_4	I_{34}	I_{234}	I
Calculated Resistance Values	R_1	R_2	R_3	R_4	R_T		
Calculated Currents by KCL	I_{34}		I_{234}		I		

QUESTIONS:

8. Suppose that in this experiment $R_1 = 4.7 \text{ k}\Omega$, $R_2 = 5.6 \text{ k}\Omega$, $R_3 = 6.8 \text{ k}\Omega$, $R_4 = 8.2 \text{ k}\Omega$ (all 5% tolerance), and $V_S = 10 \text{ V}$. The current I_{34} should be close to
 - i. 3.91 mA (b) 6.60 mA
 - (c) 3.26 mA (d) 2.69 mA
9. Suppose the conditions are as in question 1, and the following measurements are taken: $I = 6.36 \text{ mA}$, $I_1 = 2.13 \text{ mA}$, and $I_2 = 1.79 \text{ mA}$. I_{234} should be close to
 - (a) 4.23 mA (b) 2.45 mA (c) 4.48 mA (d) 4.81 mA
10. With the conditions as in question 2 and $V_S = 10 \text{ V}$, you might conclude that
 - (a) The combination R_3 and R_4 is out of tolerance
 - (b) Only R_1 is out of tolerance
 - (c) The combination R_1 and R_2 is out of tolerance
 - (d) Any of a, b, or c
11. In Figure 14-1, I_{234} is equal to
 - (a) $I - I_1$ (b) $I_2 + I_{34}$
 - (c) $I_2 + I_3 + I_4$ (d) All of these
12. Explain the total effect on the circuit if R_4 were not equal to its color-coded value

13. It is possible to have some currents less than nominal and some currents greater than nominal? How?

CONCLUSIONS:

CURRENT DIVIDERS
Experiment #12

Introduction

The experiment on "Current Dividers" is a pivotal study in electrical engineering and electronics, focusing on a circuit configuration used to distribute current among various branches in a circuit. A current divider is a parallel circuit in which the input current divides into multiple paths with inversely proportional currents based on the resistances in each path. This principle is fundamental in circuit design and analysis, especially in systems where current needs to be directed or controlled across different components.

Understanding current dividers involves not only theoretical knowledge of how currents divide in parallel circuits but also practical skills in constructing these circuits and measuring current flows. This experiment provides a hands-on approach to exploring these concepts, enhancing the understanding of current division rules and their applications in real-world circuits.

Learning Objectives

1. Understand the Principle of Current Dividers: Develop a thorough understanding of the current divider rule, its basic configuration, and how it functions in a circuit.
2. Circuit Construction and Analysis: Learn to construct current divider circuits using resistors and analyze them to predict the distribution of current across different branches.
3. Application of Ohm's Law and Parallel Circuit Principles: Apply Ohm's Law and parallel circuit principles to calculate the currents in different branches of a current divider.
4. Measurement of Current in Circuit Branches: Gain hands-on experience in measuring currents in various branches of a current divider using ammeters.
5. Understanding the Impact of Load and Resistance on Current Division: Study how different resistances and loads affect the distribution of current in a current divider circuit.
6. Analyzing Non-ideal Behaviors and Limitations: Recognize and analyze the non-ideal behaviors and practical limitations of current dividers, such as the effects of internal resistance and component tolerances.
7. Safety Practices in Circuit Experiments: Emphasize safe practices while handling and measuring electrical circuits.

8. Problem-Solving and Design Skills: Develop problem-solving skills by designing current dividers for specific requirements and troubleshooting issues in the circuit.
9. Real-World Applications: Understand the practical applications of current dividers in electronic devices and systems, enhancing the relevance of theoretical concepts.

By mastering these objectives, students will gain a solid understanding of current dividers and acquire practical skills necessary for designing and analyzing circuits where current distribution is a key factor.

EQUIPMENT:

dc power supply 0-10 V	
1 VOM or DVM	
Resistors (1/4 W):	1.6 k Ω 2.7 k Ω
	2.4 k Ω 3.3 k Ω
	± 5 percent

BACKGROUND: In experiment 11 you learned how a source voltage can be subdivided into smaller units by the use of a series resistor chain. The Voltage Divider Theorem allowed the calculation of these individual voltages without the need to calculate the current. In a similar manner, the Current Divider Theorem allows us to calculate the magnitudes of currents in a parallel circuit by a knowledge of only the total current and individual resistor values.

In this experiment you will verify the Current Divider Theorem, by adjusting for a given total current and noting the individual branch currents. Remember that resistor tolerance will affect your measured results: if only one resistor tolerance will affect you measured results: if only one resistor is out of tolerance, all resistor currents will be affected.

PROCEDURE

14. For the circuit in Figure 12-1, use the Current Divider Theorem to calculate the nominal values for the currents I_1 , I_2 , I_3 and I_4 record them in Table 12-1.
15. Construct the circuit, and adjust the source voltage until the current I is 10 mA.
16. Record the measured values of I_1 , I_2 , I_3 , and I_4 in Table 12-1 for comparison.
17. Complete the table by calculating the nominal voltage V_S to sustain the current of 10 mA. Then measure and record the actual source voltage for comparison.

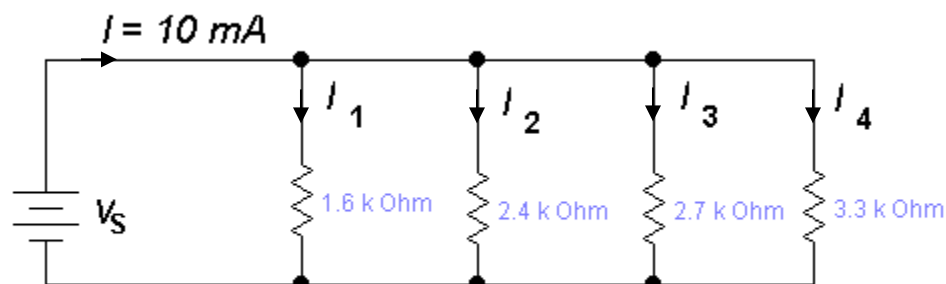


Figure 12-1

DATA

TABLE 12-1

Currents	I_1	I_2	I_3	I_4	Voltage	V_s
Nominal					Nominal	
Measured					Measured	

QUESTIONS:

- If all the resistance values in Figure 15-1 were stepped up by a factor of 10 (and I remained constant at 10 mA), the new currents would be
 - 10 times their old values
 - (b) Unchanged
 - (c) One-tenth their old values
- When resistor tolerance is taken into account, the smallest expected value for I_3 in Figure 15-1
 - (a) 2.16 mA (b) 1.99 mA (c) 2.08 mA (d) None of these
- If the 2.4 k Ω resistor in Figure 15-1 were mistakenly replaced with a 24 k Ω resistor and I_T were maintained constant, the currents affected would be
 - (a) I_1 , I_2 , I_3 , and I_4 (b) I_2 , I_3 , and I_4
 - (c) I_1 , I_3 , and I_4 (d) I_2 only
- If all the resistors in Figure 15-1 were at the upper boundaries of their tolerance, the voltage V_s required to sustain 10 mA through the combination would be
 - (a) 5.83 V (b) 6.12 V (c) 5.54 V (d) None of these
- The smallest valued resistance in a parallel connection of resistors will always take the largest current. Explain why this is so.

6. If the current I in Figure 15-1 is kept constant, comment on the effect of varying any single resistor on the branch currents I_1 , I_2 , I_3 , and I_4 .

CONCLUSIONS:

Introduction to Electronic Workbench
Experiment # 13

Introduction

The experiment on using an Electronic Workbench, specifically software tools like Multisim and Proteus, is an essential part of modern electrical engineering and electronics education. These software platforms are powerful tools for simulating and analyzing electronic circuits without the need for physical components or the risks associated with live experimentation. Multisim and Proteus offer virtual environments where users can design, test, and visualize the behavior of electronic circuits, providing an invaluable resource for learning, prototyping, and troubleshooting.

These tools simulate a wide range of electronic components and their interactions, allowing students to experiment with complex circuit designs and understand their functionality before actual implementation. The experiment typically involves creating circuit designs, running simulations to analyze circuit behavior, and using the software's analytical tools to understand various aspects of electronic circuits.

Learning Objectives

- a. Familiarization with Electronic Workbench Software: Gain proficiency in using Multisim, Proteus, or similar electronic workbench software, including understanding their user interfaces and features.
- b. Circuit Design and Simulation: Learn to design various types of electronic circuits using the software, including both analog and digital circuits, and run simulations to analyze their behavior.
- c. Component Selection and Utilization: Understand how to select and utilize different electronic components within the software, including resistors, capacitors, inductors, transistors, and integrated circuits.
- d. Analysis of Circuit Behavior: Develop skills in analyzing the behavior of circuits through simulations, including understanding voltage, current, frequency responses, and other important parameters.
- e. Troubleshooting and Optimization: Use the software to troubleshoot and optimize circuit designs, identifying and correcting design flaws or inefficiencies.
- f. Comparison of Theoretical and Simulated Results: Learn to compare theoretical calculations and expected behaviors with the results obtained from simulations, understanding the limitations and accuracies of simulations.
- g. Understanding of Advanced Features: Explore advanced features of the software like PCB layout design, 3D visualization, and SPICE simulations, enhancing the depth of circuit analysis and design capabilities.
- h. Safety and Practical Application Awareness: While working virtually, understand the importance of safety and practical considerations in real-

world circuit design and implementation.

- i. Real-World Problem-Solving Skills: Develop real-world problem-solving skills by designing circuits that address specific tasks or challenges, simulating real-life scenarios.

By mastering these objectives, students will not only gain a comprehensive understanding of electronic circuit design and analysis but also acquire valuable skills in using modern simulation tools, which are crucial for anyone pursuing a career in electrical engineering and electronics.

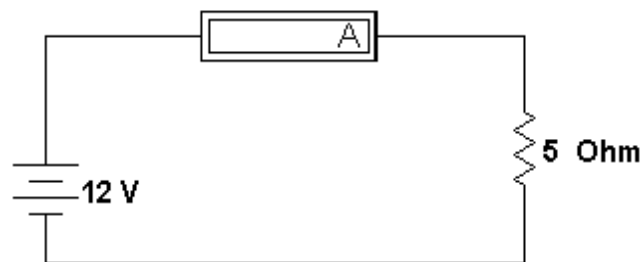
Materials Required

1 set Personal Computer with electronic workbench

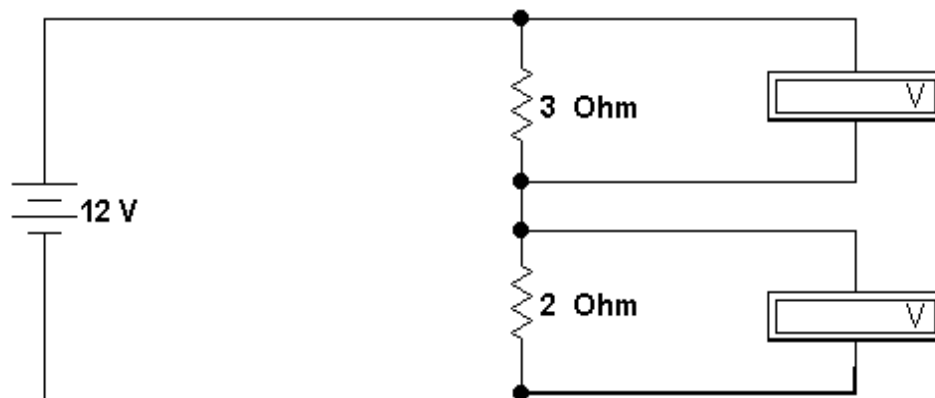
Procedure

Create the following diagram using the electronic workbench

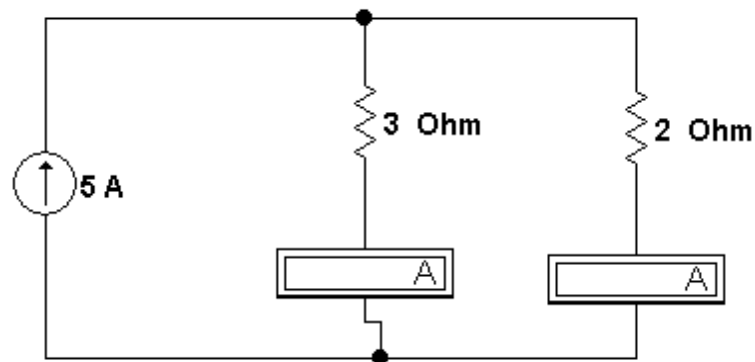
4. Ohm's Law.



5. Voltage Division



6. Current Division



7. What advantages might an analog meter have over a digital meter in measuring voltages and current in a laboratory experiment?

SUPERPOSITION Experiment 14

Introduction

The experiment on "Superposition" is a fundamental study in the field of electrical engineering and electronics, focusing on a key principle used in the analysis of linear circuits. The superposition theorem states that in a linear circuit with multiple independent sources (voltage or current), the voltage or current at any component in the circuit is the algebraic sum of the voltages or currents caused by each independent source acting alone, while all other independent sources are replaced by their internal impedances. This theorem simplifies the analysis of complex circuits by allowing the study of each independent source's effect in isolation.

The superposition experiment typically involves constructing linear circuits with multiple independent sources, then systematically analyzing the contribution of each source to the circuit's overall behavior. This approach not only reinforces the understanding of linear circuit theory but also enhances analytical and problem-solving skills.

Learning Objectives

1. Understanding the Superposition Theorem: Develop a solid understanding of the superposition theorem, its scope, and limitations, particularly its applicability only to linear circuits.
2. Circuit Construction for Superposition Analysis: Gain practical experience in constructing circuits with multiple independent sources suitable for applying the superposition theorem.
3. Analysis of Individual Source Effects: Learn to analyze the effect of each independent source on the circuit, including calculating voltages and currents when each source acts alone.
4. Measurement and Calculation Techniques: Develop skills in accurately measuring and calculating voltages and currents in circuits, and understand the importance of linear superposition in these measurements.
5. Algebraic Summation of Effects: Enhance the ability to algebraically sum the individual effects of each source to determine the overall response of the circuit.
6. Problem-Solving and Critical Thinking: Foster problem-solving and critical thinking skills by dissecting complex circuit behaviors into simpler, manageable parts using the superposition principle.

7. **Safety in Circuit Experiments:** Emphasize safe practices in handling and constructing electrical circuits.
8. **Real-World Application and Limitations:** Understand the practical applications of the superposition theorem in real-world scenarios and recognize its limitations, especially in non-linear circuits.
9. **Data Analysis and Interpretation:** Develop proficiency in interpreting experimental data, comparing it with theoretical predictions, and understanding the practical considerations in applying superposition.

By mastering these objectives, students will not only learn a crucial analytical tool in circuit analysis but also acquire practical skills that are essential for the design and troubleshooting of complex electrical systems.

EQUIPMENT:

- 2 dc power supply 0-12 V
- 1 VOM or DVM
- Resistors (1/4 W):
 - 1 k Ω
 - 2 k Ω
 - 3 k Ω
 - ± 5 percent

BACKGROUND: The Superposition Principle allows us to calculate the combined effects of a multisource circuit by looking at the individual effects of each source acting alone, then summing them. Particular attention must be paid to current directions and voltage polarities when applying superposition. The experiment uses a circuit with two voltage sources, V_{S1} , and V_{S2} (see Figure 22-1). The three currents I_1 , I_2 , and I_3 can be calculated by taking the components due to V_{S1} acting alone, then V_{S2} acting alone, and adding them together. You will recall that the source not being considered is to be killed or replaced with its internal resistance. This is the same as replacing it with a short circuit.

It is important to note that you must remove the power supply (or switch it off) before you replace it with a short: never short-circuit a functioning power supply under these circumstances. Also, pay attention to the directions of the current components due to each source so that you *algebraically* sum them to get the totals.

PROCEDURE

1. For the circuit in Figure 22-1, use the Superposition Principle to calculate the components of I_1 , I_2 and I_3 due to each source acting alone, and record your results in the table of calculated data, Table 22-1. Note that I_{11} is the component of I_1 due to V_{S1} , I_{12} is the component of I_1 due to V_{S2} , etc.
2. Complete Table 22-1, by calculating the actual values of I_1 , I_2 , and I_3 when both V_{S1} and V_{S2} are active.

3. Connect the circuit of Figure 22-1. Remove power supply V_{S1} and replace it with a short circuit. Then measure the currents I_{11} , I_{21} , and I_{31} and record them in the table of measured data, Table 22-2, in the appropriate column.
4. Repeat step 3, while replacing V_{S1} with a short circuit. This will enable you to measure and record I_{12} , I_{22} , and I_{32} .
5. Finally connect the complete circuit with both sources active. Measure and record the actual currents I_1 , I_2 , and I_3 . Compare your measured and calculated data.

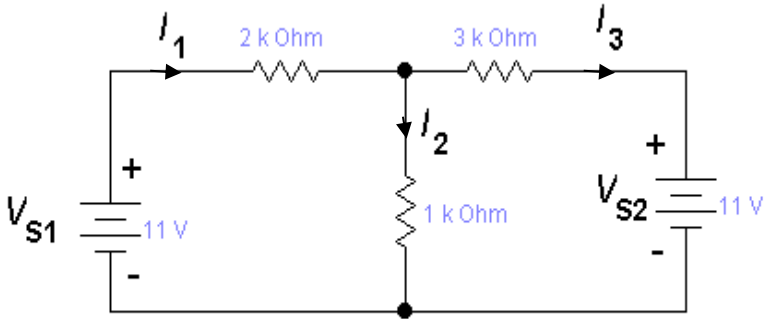


FIGURE 22-1

TABLE 22-1

Calculate Data					
V_{S1} Active		V_{S2} Active		V_{S1} and V_{S2} Active (algebraic sum)	
I_{11}		I_{12}		I_1	
I_{21}		I_{22}		I_2	
I_{31}		I_{32}		I_3	

TABLE 22-2

Measured Data					
V_{S1} Active		V_{S2} Active		V_{S1} and V_{S2} Active (algebraic sum)	
I_{11}		I_{12}		I_1	
I_{21}		I_{22}		I_2	
I_{31}		I_{32}		I_3	

NOTES

QUESTIONS FOR EXPERIMENT 14

1. With element values as in Figure 22-1, if the source V_{S1} were made equal to 22 V, then
 - i. I_3 would double
 - (b) I_3 would have the same magnitude, but be opposite in direction
 - (c) I_{12} , I_{22} , and I_{32} would each double
 - (d) I_{11} , I_{21} , and I_{31} would each double
2. The direction of I_2 is downward and independent of the *magnitudes* of V_{S1} and V_{S2} .
 - (a) True
 - (b) FalseIf both V_{S1} and V_{S2} are increased by a factor of two, then
 - (a) All currents will double
 - (b) I_2 only will double
 - (c) All currents will remain constant
 - (d) I_1 and I_3 will double and I_2 will stay the same
3. The current I_3 depends on
 - (a) The value of V_{S2}
 - (b) The values of V_{S2} , R_3 , and R_2
 - (c) The values of V_{S1} and R_1
 - (d) All of the above
4. Explain why the Superposition Principle does not work for power in the elements of a circuit.
5. With values as in Figure 22-1, calculate the value to which V_{S1} must be reduced in order to force the current I_1 to equal zero.

Thevenin's Theorem
Experiment # 15

Introduction

The experiment on "Thevenin's Theorem" is an important component in electrical engineering education, focusing on a fundamental network theorem that simplifies the analysis of complex electrical circuits. Thevenin's Theorem states that any linear, bilateral electrical network with voltage and current sources and resistances can be simplified to a single voltage source in series with a single resistance. This theorem is instrumental in analyzing circuits by reducing them to simpler forms, thus making calculations more manageable and the understanding of circuit behavior more intuitive.

The experiment typically involves taking a complex circuit with multiple components and reducing it to its Thevenin equivalent. This process includes identifying the Thevenin equivalent voltage (V_{th}) and Thevenin equivalent resistance (R_{th}) of the original circuit. Students gain practical experience in circuit analysis and a deeper understanding of how to apply Thevenin's Theorem in various electrical network scenarios.

Learning Objectives

- a. Understand Thevenin's Theorem: Gain a comprehensive understanding of Thevenin's Theorem, including its statement, conditions of applicability, and the concept of linear and bilateral networks.
- b. Circuit Analysis and Reduction: Learn to analyze complex circuits and simplify them to their Thevenin equivalents, focusing on identifying and calculating the Thevenin voltage and resistance.
- c. Practical Application of Theorem: Gain hands-on experience in applying Thevenin's Theorem to real-world circuit problems, enhancing the understanding of its practical utility.
- d. Measurement and Calculation Skills: Develop skills in measuring electrical parameters and calculating the Thevenin equivalent of given circuits using theoretical analysis and practical measurements.
- e. Problem-Solving and Critical Thinking: Enhance problem-solving and critical-thinking abilities by tackling various circuit configurations and determining their Thevenin equivalents.
- f. Understanding the Impact of Load on Circuits: Study the impact of connecting a load to the Thevenin equivalent circuit and how it relates to the behavior of the original circuit.
- g. Safety in Electrical Experiments: Emphasize the importance of safety practices while handling electrical components and equipment during experimentation.
- h. Data Analysis and Interpretation: Cultivate the ability to interpret experimental data, compare it with theoretical calculations, and understand discrepancies and practical limitations.

- i. Application in Circuit Design and Troubleshooting: Understand the application of Thevenin's Theorem in simplifying the process of circuit design and troubleshooting.

By mastering these objectives, students will not only comprehend the theoretical aspects of Thevenin's Theorem but also acquire practical skills essential for effective circuit analysis, design, and troubleshooting in various electrical engineering applications. Materials Required

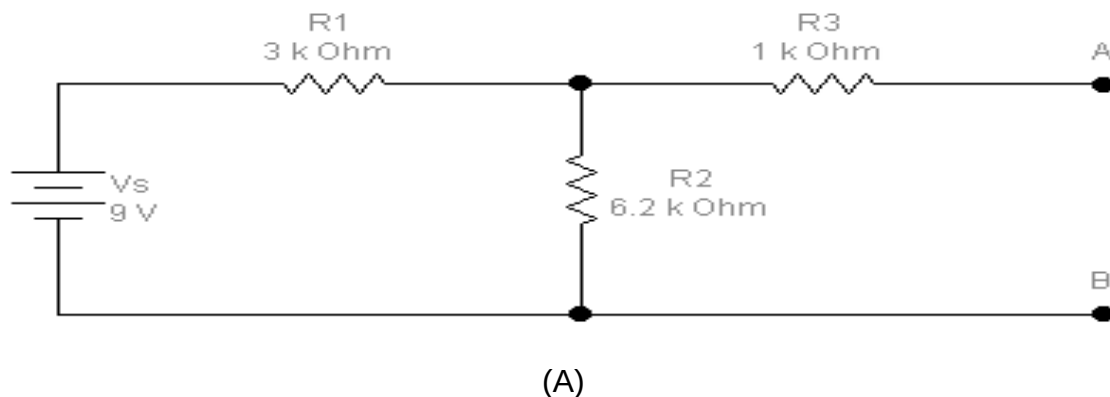
Power Supply: 1 Variable 0-15 V dc, regulated

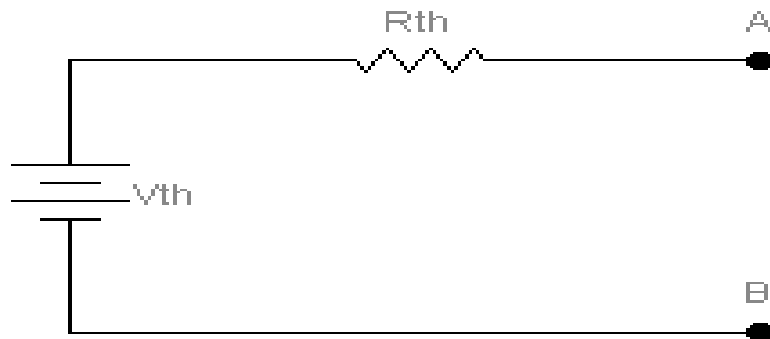
Instruments: 1 Analog multimeter

Resistors:
1 1 k Ω
2 3 k Ω
1 6.2 k Ω

Background

Thevenin's Theorem allows us to replace any circuitry (no matter how complex) "behind" two chosen terminals by a simple voltage source series resistor equivalent. To demonstrate this idea we will apply it to the simple three-resistor circuit of the figure. According to the theorem, we should be able to replace the circuitry to the left of the terminals A – B in the figure A with that in figure B. To obtain the equivalent source voltage, you will measure the open circuit voltage. The Thevenin resistance is obtained by measuring R_{A-B} with a "dead source" replaced with a short circuit). To verify that this combination is indeed equivalent, you will then connect a load to both circuits, and verify that the resulting voltage and current are the same in both cases. Once again, do not simply short your power supply - remove it, and replace it with a piece of wire.





(B)

Procedure

1. For the circuit in Figure A, use Thevenin's Theorem to calculate the values of V_{TH} and R_{TH} , and record them in Table A.
2. Measure the open circuit voltage V_{AB} and record this V_{TH} under the measured column of Table A.
3. Replace the source with a short circuit, and measure the resistance between the terminals A and B. Record this as R_{TH} under the measured column of Table A.
4. Calculate the voltage across and current through a $3\text{ K}\Omega$ load that is to be placed across the terminals A and B. Perform the calculations for both the actual circuit and its Thevenin equivalent. The results should be identical. Record the results in the column headed "Calculated" in Table B.
5. Connect a $3\text{ K}\Omega$ load to the terminals A and B of the circuit in Figure A. Measure the resulting load current and voltage, and record them in Table B.
6. Construct the circuit of Figure B with the calculated values of V_{TH} and R_{TH} . Connect a $3\text{ K}\Omega$ load to the terminals A and B. Measure the resulting load current and voltage, and record them in Table B. They should agree closely with those in the adjacent columns.

Table A

Thevenin Parameters			
Calculated		Measured	
V_{TH}		V_{TH}	
R_{TH}		R_{TH}	

Table B

Loaded Circuit Parameters					
Calculated		Measured			
		Actual Circuit		Thevenin Equivalent	
I_L		I_L		I_L	
V_L		V_L		V_L	

Questions

1. In figure A, the Thevenin resistance R_{TH} depends on

- a. V_s
- b. R_1 , R_2 , R_3 , and V_s
- c. R_1 and R_2 only
- d. R_1 , R_2 , and R_3

2. In figure A, the Thevenin voltage V_{TH} depends on

- a. V_s , R_1 , R_2 , and R_3
- b. V_s only
- c. V_s , R_1 , and R_2
- d. V_s and R_1 only

3. If R_3 were increased in value, the Thevenin voltage would

- a. Increase
- b. Decrease
- c. Remain the same
- d. insufficient information

4. In addition resistance placed across the terminals A and B (before any load is connected) would

- a. Increase R_{TH}
- b. Decrease R_{TH}
- c. Not change R_{TH}
- d. insufficient information

5. A resistor placed directly in parallel with the source voltage V_s does not affect R_{TH} . Why?

6. The Thevenin equivalent resistance R_{TH} for the network in figure A was 3 K Ω . Detail how this could be altered to 2 K Ω by using a single resistor placed across the terminals A and B. Calculate the value of the resistor that will accomplish this. Will the Thevenin voltage change?

MAXIMUM POWER TRANSFER Experiment 15

Introduction

The experiment on "Maximum Power Transfer" is a crucial aspect of electrical engineering and electronics, focusing on a key principle for optimizing the efficiency of power transfer in circuits. The Maximum Power Transfer Theorem states that a source will deliver maximum power to a load when the load resistance is equal to the source resistance (or the internal resistance of the source). This concept is of paramount importance in designing systems for efficient energy transfer, such as in audio engineering, telecommunications, and power electronics.

The experiment typically involves setting up a circuit with an adjustable load resistor and measuring the power transferred to the load under various conditions. By varying the load resistance and observing the effects on power transfer, students gain a practical understanding of the theorem and its implications in real-world applications.

Learning Objectives

1. Understanding the Maximum Power Transfer Theorem: Develop a comprehensive understanding of the theorem, including its statement, conditions, and theoretical background.
2. Circuit Setup for Experimentation: Learn to set up a circuit that can be used to demonstrate and explore the Maximum Power Transfer Theorem, including the use of variable resistors as loads.
3. Measurement of Power Transfer: Gain hands-on experience in measuring voltage, current, and consequently power in a circuit, and how these measurements change with varying load resistance.
4. Analyzing the Relationship Between Load and Source Resistance: Understand the relationship between load resistance, source resistance, and the power transferred to the load.
5. Data Analysis and Graphical Representation: Develop skills in analyzing experimental data, plotting graphs of power versus load resistance, and identifying the point of maximum power transfer.
6. Safety in Electrical Experiments: Emphasize safe practices in handling electrical circuits, especially when dealing with power measurements.
7. Critical Thinking and Problem-Solving: Enhance critical thinking and problem-solving skills by applying the theorem to various circuit configurations and real-world scenarios.

8. Understanding Practical Limitations: Recognize the practical limitations and considerations when applying the Maximum Power Transfer Theorem in real-world applications, such as efficiency and heat dissipation.
9. Application in System Design: Understand the implications of the theorem in the design of electrical and electronic systems, particularly where power efficiency is crucial.

By mastering these objectives, students will not only learn the fundamental principles of maximum power transfer but also acquire practical skills and insights necessary for designing and analyzing efficient power systems in various fields of electrical and electronic engineering.

EQUIPMENT:

dc power supply 0-10 V

1 VOM or DVM

Resistors (1/4 W):

1 k Ω	3.3 k Ω
1.5 k Ω	4.3 k Ω
2 k Ω	5.6 k Ω
2.7 k Ω	7.5 k Ω
3 k Ω (two)	10 k Ω
	± 5 percent

BACKGROUND: In this experiment you are going to answer the question, “Given a voltage source with some fixed internal resistance R_S , what value of load resistor will absorb the most power from the circuit?” Clearly, there are an infinite number of loads from $0\ \Omega$ (a short circuit) to $\infty\ \Omega$ (an open circuit). Since a short circuit would have not voltage and an open circuit no current, both of these produce zero load power. The answer lies somewhere between $0\ \Omega$ and $\infty\ \Omega$. The maximum power transfer theorem tells us that the load should be equal in magnitude to the source resistance for maximum power to be absorbed by the load.

In the experiment you will connect a series of increasing load resistors to a source resistor combination. In each case, you will measure V_L and I_L and calculate $P_L (= I_L V_L)$, the power in the load. You should find that this power is maximized where $R_L = R_S$. A graph is one of the best ways to illustrate the theorem. Be sure to join your data points with a smooth curve (not straight line segments) so that the true variation of power with load resistance can be observed.

PROCEDURE

- Given $V_S = 10\text{ V}$ and $R_S = 3\text{ k}\Omega$, calculate the load voltage V_L , load current I_L , and load power P_L for each value of R_L in Table 24-1.
- Set up the source with $V_S = 10\text{ V}$ and $R_S = 3\text{ k}\Omega$. For each value of R_L , measure and record the load voltage V_L and the load current I_L . From this, calculate the load power P_L . complete Table 24-2 for measured data.
- Plot a graph of load power P_L versus load resistance R_L from your measured data. Where is the load power a maximum and what is its value?

DATA FOR EXPERIMENT 15

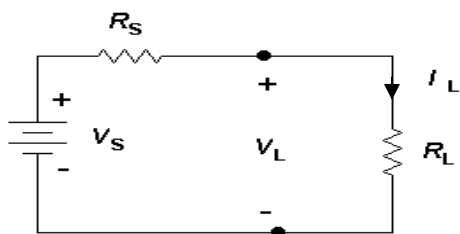


FIGURE 24-1

TABLE 24-1

Calculate Data										
R_L (k Ω)	1	1.5	2	2.7	3	3.3	4.3	5.6	7.5	10
V_L (V)										
I_L (mA)										
P_L (mW)										

TABLE 22-2

Measured Data										
R_L (k Ω)	1	1.5	2	2.7	3	3.3	4.3	5.6	7.5	10
V_L (V)										
I_L (mA)										
P_L (mW)										

1. The load power is maximum when
 - i. $R_L \ll R_S$ (b) $R_L \gg R_S$ (c) $R_L = R_S$
2. If the source resistance were made larger than $3\text{ k}\Omega$ in this experiment, then
 - (a) All the values of P_L would be less
 - (b) All the values of P_L would be greater
 - (c) The values of P_L would not change
 - (d) Some values of P_L would be smaller and some greater
3. The power dissipated in the source resistance depends on the value of the load resistance.
 - (a) $\leq P_L$ (b) $< P_L$ (c) $\geq P_L$ (d) None of these
4. The power dissipated in the source resistance depends on the value of the load resistance.
 - (a) True (b) False
5. Explain qualitatively why the P_L versus R_L curve is not symmetrical about $R_L = R_S$.
6. Show that the efficiency of the source is only 50 percent when supplying its maximum power.

CAPACITOR CHARGING AND DISCHARGING

Experiment # 17

Introduction

The experiment on "Capacitor Charging and Discharging" is a fundamental aspect of learning in electrical engineering and physics, focusing on the behavior of capacitors in circuits. Capacitors, known for their ability to store and release electrical energy, charge up when connected to a power source and discharge when the source is removed. This experiment is crucial for understanding the dynamic behavior of capacitors, characterized by the time it takes to charge and discharge, which is influenced by the circuit's resistance and the capacitor's capacitance.

This experiment typically involves setting up circuits with capacitors and resistors, and using instruments like voltmeters and oscilloscopes to observe and measure the charging and discharging processes. The investigation provides insights into important concepts such as time constants and exponential functions that describe how capacitors charge and discharge over time.

Learning Objectives

1. **Understand Capacitor Function and Behavior:** Gain a thorough understanding of how capacitors work, their function in circuits, and the basic principles governing their charging and discharging processes.
2. **Circuit Construction and Analysis:** Learn to construct circuits for studying capacitor charging and discharging, including the correct setup of capacitors, resistors, and power supplies.
3. **Measurement of Voltage and Time Constants:** Develop skills in measuring voltage across a capacitor over time using voltmeters or oscilloscopes, and understand how to determine the time constant of the circuit.
4. **Application of Theoretical Principles:** Apply theoretical principles such as the RC time constant formula to predict and verify the behavior of capacitors in charging and discharging circuits.
5. **Graphical Analysis of Charging and Discharging Curves:** Enhance the ability to plot and analyze the charging and discharging curves of capacitors, understanding the exponential nature of these processes.

6. Safety in Electrical Experiments: Emphasize safe handling practices when working with electrical circuits, especially when dealing with stored energy in capacitors.
7. Interpreting Experimental Data: Develop skills in interpreting experimental data, comparing it with theoretical predictions, and understanding any discrepancies.
8. Problem-Solving and Critical Thinking: Foster problem-solving and critical-thinking skills by troubleshooting circuit issues and exploring the impact of different circuit components on the charging/discharging process.
9. Real-World Applications of Capacitors: Understand the practical applications of capacitors in various electronic devices and systems, highlighting the importance of their charging and discharging characteristics.

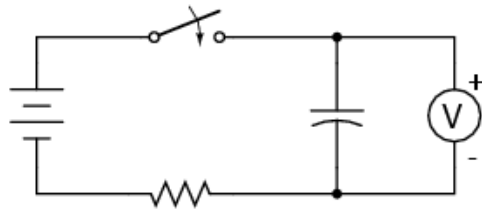
By mastering these objectives, students will not only comprehend the fundamental properties and behavior of capacitors but also acquire practical skills in circuit analysis and experimentation, which are essential for a wide range of applications in electrical and electronic engineering.

- 6 volt battery
- Two large electrolytic capacitors, 1000 μF minimum
- Two 1 k Ω resistors
- One toggle switch, SPST ("Single-Pole, Single-Throw")

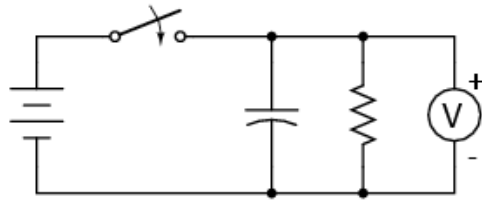
Large-value capacitors are required for this experiment to produce time constants slow enough to track with a voltmeter and stopwatch. Be warned that most large capacitors are of the "electrolytic" type, and they are *polarity sensitive*! One terminal of each capacitor should be marked with a definite polarity sign. Usually capacitors of the size specified have a negative (-) marking or series of negative markings pointing toward the negative terminal. Very large capacitors are often polarity-labeled by a positive (+) marking next to one terminal. Failure to heed proper polarity will almost surely result in capacitor failure, even with a source voltage as low as 6 volts. When electrolytic capacitors fail, they typically explode, spewing caustic chemicals and emitting foul odors. Please, try to avoid this!

I recommend a household light switch for the "SPST toggle switch" specified in the parts list.

SCHEMATIC DIAGRAM

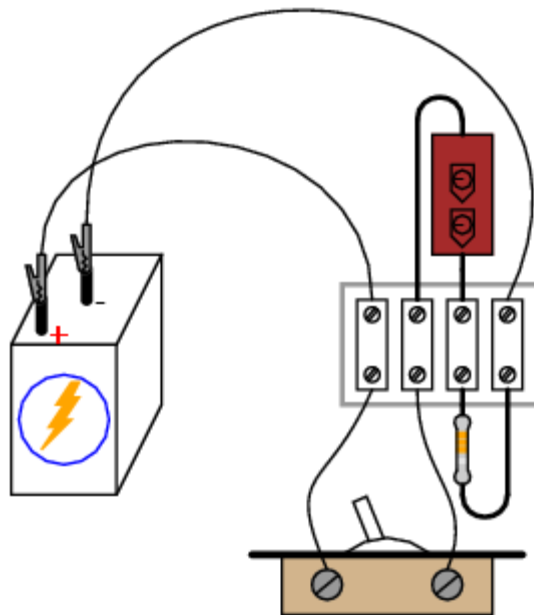


Charging circuit

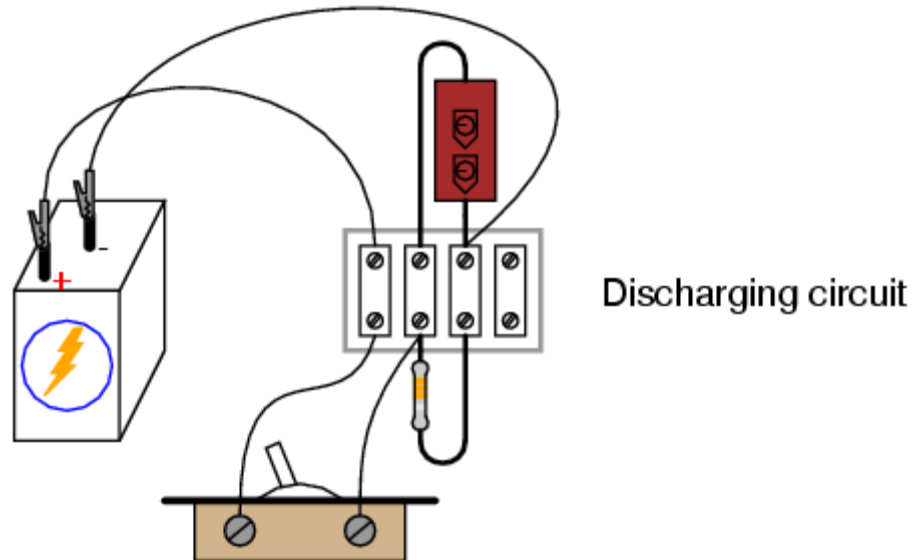


Discharging circuit

ILLUSTRATION



Charging circuit



INSTRUCTIONS

Build the "charging" circuit and measure voltage across the capacitor when the switch is closed. Notice how it increases slowly over time, rather than suddenly as would be the case with a resistor. You can "reset" the capacitor back to a voltage of zero by shorting across its terminals with a piece of wire.

The "time constant" (τ) of a resistor capacitor circuit is calculated by taking the circuit resistance and multiplying it by the circuit capacitance. For a $1\text{ k}\Omega$ resistor and a $1000\text{ }\mu\text{F}$ capacitor, the time constant should be 1 second. This is the amount of time it takes for the capacitor voltage to increase approximately 63.2% from its present value to its final value: the voltage of the battery.

It is educational to plot the voltage of a charging capacitor over time on a sheet of graph paper, to see how the inverse exponential curve develops. In order to plot the action of this circuit, though, we must find a way of slowing it down. A one-second time constant doesn't provide much time to take voltmeter readings!

We can increase this circuit's time constant two different ways: changing the total circuit resistance, and/or changing the total circuit capacitance. Given a pair of identical resistors and a pair of identical capacitors, experiment with various series and parallel combinations to obtain the slowest charging action. You should already know by now how multiple resistors need to be connected to form a greater total resistance, but what about capacitors? This circuit will demonstrate to you how capacitance changes with series and parallel capacitor connections. Just be sure that you insert the capacitor(s) in the proper direction: with the ends labeled negative (-) electrically "closest" to the battery's negative terminal!

The discharging circuit provides the same kind of changing capacitor voltage, except this time the voltage jumps to full battery voltage when the switch closes and slowly falls when the switch is opened. Experiment once again with different

combinations of resistors and capacitors, making sure as always that the capacitor's polarity is correct.

RATE-OF-CHANGE INDICATOR

Experiment # 18

Introduction

The experiment on the "Rate-of-Change Indicator" is an intriguing and educational exploration in the fields of physics, electronics, and engineering. This experiment is centered around creating a device or system that can detect and indicate the rate at which a particular physical quantity, such as temperature, light intensity, or velocity, changes over time. The principle behind a rate-of-change indicator is often based on measuring the derivative of a signal with respect to time, which provides insights into how quickly the monitored parameter is varying.

Such an experiment is particularly relevant in a world where dynamic systems are increasingly common, and understanding the rate of change of various parameters is crucial for control systems, predictive maintenance, and real-time monitoring applications. Students will engage in designing a circuit or using sensors and microcontrollers to construct a system that can effectively measure and display the rate of change of a chosen physical quantity.

Learning Objectives

1. Understanding the Concept of Rate of Change: Gain a solid understanding of the concept of rate of change in physical systems and its mathematical and practical implications.
2. Circuit Design and Sensor Integration: Learn to design and construct a circuit or system that can measure the rate of change of a chosen parameter, incorporating appropriate sensors and possibly microcontrollers.
3. Signal Processing and Analysis: Develop skills in processing and analyzing the signal obtained from the sensor to determine the rate of change, which may involve understanding and implementing derivative calculations.
4. Data Display and Interpretation: Learn to effectively display the rate-of-change data, possibly using LEDs, digital displays, or graph plotting, and interpret what this data signifies in the context of the chosen physical parameter.
5. Practical Application of Electronics and Programming: Apply electronic skills and, if relevant, programming skills for

microcontroller-based systems to create a functional rate-of-change indicator.

6. Experimentation and Calibration: Engage in experimentation and calibration processes to ensure accurate and reliable measurements of the rate of change.
7. Safety and Best Practices: Emphasize safety and best practices in handling electronic components, sensors, and any other involved equipment.
8. Problem-Solving and Innovative Thinking: Encourage problem-solving and innovative thinking in designing a system that effectively measures and indicates the rate of change, considering various design challenges and constraints.
9. Real-World Applications: Understand the real-world applications of rate-of-change indicators in various fields, such as environmental monitoring, industrial processes, and automotive systems.

By completing this experiment, students will not only gain a deeper understanding of the concept of rate of change but also acquire practical skills in sensor integration, signal processing, and system design, which are highly applicable in the modern technological landscape.

PARTS AND MATERIALS

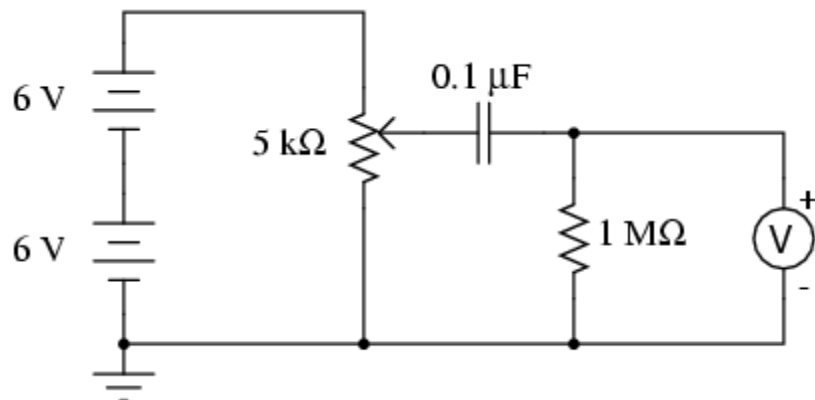
- Two 6 volt batteries
- Capacitor, 0.1 μF
- 1 M Ω resistor
- Potentiometer, single turn, 5 k Ω , linear taper

The potentiometer value is not especially critical, although lower-resistance units will, in theory, work better for this experiment than high-resistance units. I've used a 10 k Ω potentiometer for this circuit with excellent results.

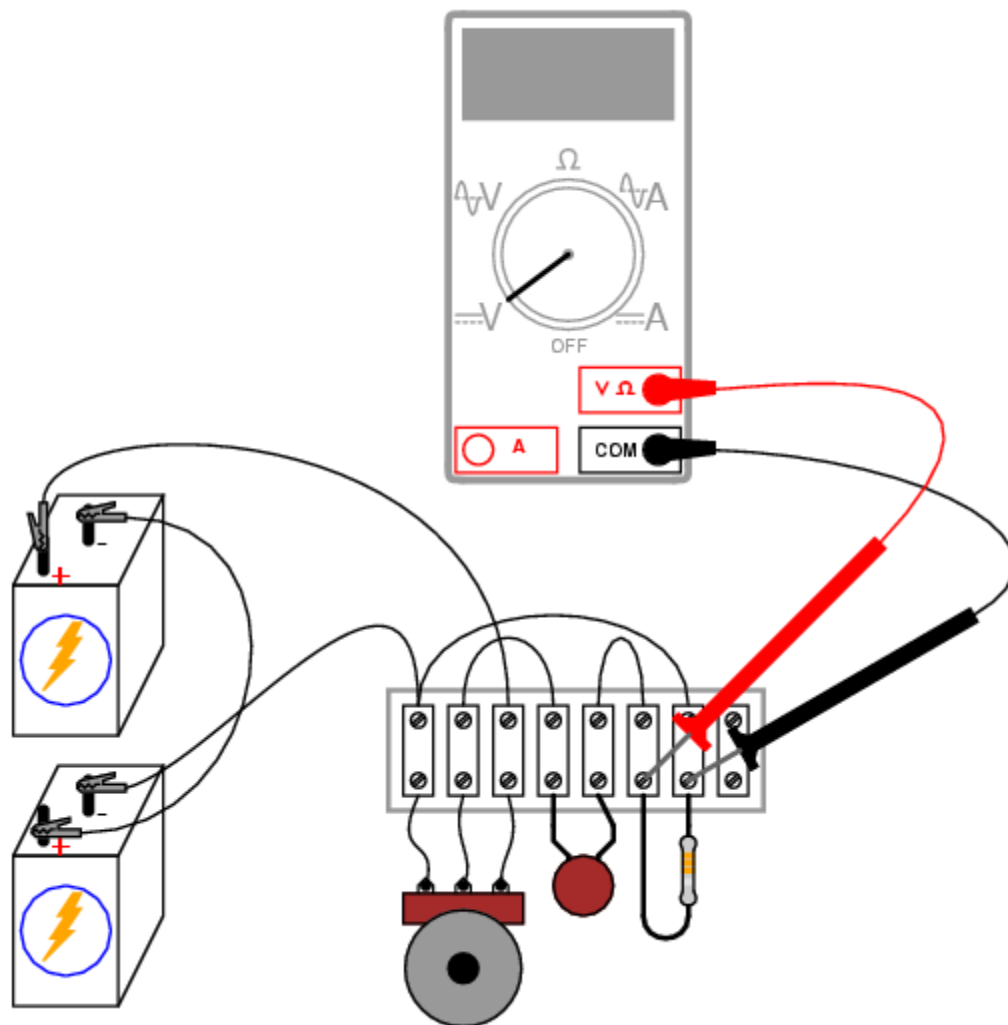
LEARNING OBJECTIVES

- How to build a differentiator circuit
- Obtain an empirical understanding of the *derivative* calculus function

SCHEMATIC DIAGRAM



ILLUSTRATION



INSTRUCTIONS

Measure voltage between the potentiometer's wiper terminal and the "ground" point shown in the schematic diagram (the negative terminal of the lower 6-volt battery). This is the input voltage for the circuit, and you can see how it smoothly varies between zero and 12 volts as the potentiometer control is turned full-range. Since the potentiometer is used here as a voltage divider, this behavior should be unsurprising to you.

Now, measure voltage across the 1 M Ω resistor while moving the potentiometer control. A digital voltmeter is highly recommended, and I advise setting it to a very sensitive (millivolt) range to obtain the strongest indications. What does the voltmeter indicate while the potentiometer is *not* being moved? Turn the potentiometer slowly clockwise and note the voltmeter's indication. Turn the potentiometer slowly counter-clockwise and note the voltmeter's indication. What difference do you see between the two different directions of potentiometer control motion?

Try moving the potentiometer in such a way that the voltmeter gives a steady, small indication. What kind of potentiometer motion provides the *steadiest* voltage across the 1 M Ω resistor?

In calculus, a function representing the rate of change of one variable as compared to another is called the *derivative*. This simple circuit illustrates the concept of the derivative by producing an output voltage proportional to the input voltage's *rate of change over time*. Because this circuit performs the calculus function of differentiation with respect to time (outputting the time-derivative of an incoming signal), it is called a *differentiator* circuit.

Like the *average* circuit shown earlier in this chapter, the differentiator circuit is a kind of analog computer. Differentiation is a far more complex mathematical function than averaging, especially when implemented in a digital computer, so this circuit is an excellent demonstration of the elegance of analog circuitry in performing mathematical computations.

More accurate differentiator circuits may be built by combining resistor-capacitor networks with electronic *amplifier* circuits. For more detail on computational circuitry, go to the "Analog Integrated Circuits" chapter in this Experiments volume.